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碩士學位論文

六軸工業機械手臂的線材組裝姿態補償

Six-Axis Industrial Manipulators Pose
Compensation For Cable Assembly

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摘要

在現代工業自動化中，機械手臂已廣泛應用於高重複性與高精度之組裝任務。然而，當操作對象由剛性零件轉變為柔性線性物體 (Flexible Linear Object, FLO)，例如線材或纜線時，傳統以位置控制為主的自動化方法將面臨明顯挑戰。由於線材容易受到重力、夾持姿態與材料柔性影響而產生變形，實際線端位置可能會偏離理想工具中心點 (Tool Center Point, TCP)，進而降低精密插入任務之成功率。此外，多數商用工業機械手臂控制器屬於封閉式「黑盒」系統，使用者難以取得其內部運動學演算法，也不易針對特定任務建立補償策略。

本論文提出一套用於機械手臂線材插入任務之模型導向姿態補償方法。研究中以六軸工業機械手臂 (ABB IRB 1090) 為對象，並透過 Modified Denavit-Hartenberg (MDH) 方法重新建立其外部運動學模型，使該模型能獨立於商用控制器之外運作。基於此模型，本研究可數學化描述機械手臂法蘭、夾爪、理想 TCP、彎曲後線端與目標插入點之間的空間關係。所提出方法主要針對線材夾持後因彎曲所造成的角度偏差進行補償。此方法並不直接改變實體機械手臂的 TCP，而是在模擬環境中計算虛擬補償 TCP，並透過逆運動學求得對應之各軸角度。

為驗證此方法之可行性，本研究使用 MATLAB 與 GeoGebra 3D 模擬線端姿態補償過程，並以 1.15 mm 公差之精密插入任務作為驗證案例。接著，將模擬模型所產生之關節角度與實體機械手臂操作結果進行比較與套用。結果顯示，基於 MDH 之姿態補償模型可使彎曲線端對準插入目標，並提升商用機械手臂系統執行全自動線材插入任務之可行性。

Abstract

In modern industrial automation, manipulators have been widely applied to repetitive and high-precision assembly tasks. However, when the target object changes from a rigid component to a Flexible Linear Object (FLO), such as a wire or cable, conventional position-based automation methods face significant challenges. Because wires are easily deformed by gravity, gripping posture, and material flexibility, the actual wire tip may deviate from the ideal tool center point (TCP), reducing the success rate of precision insertion. In addition, most commercial industrial manipulators operate through closed “black-box” controllers, making it difficult for users to access internal kinematic algorithms or develop task-specific compensation strategies.

This thesis proposes a model-oriented pose compensation method for manipulator wire insertion using a six-axis industrial manipulator. The ABB IRB 1090 manipulator is remodeled through the Modified Denavit-Hartenberg (MDH) convention to establish an external kinematic model independent of the commercial controller. Based on this model, the relationship among the manipulator flange, gripper, ideal TCP, bent wire tip, and target insertion point can be mathematically described. The proposed method focuses on compensating for the angular deviation caused by wire bending after grasping. Instead of physically redefining the manipulator TCP, an imaginary compensated TCP is calculated in the simulation environment, allowing the manipulator joint angles to be obtained through inverse kinematics.

To verify the feasibility of the proposed framework, MATLAB and Ge-

oGebra 3D were used to simulate the wire-tip compensation process under a precision insertion task with a 1.15 mm tolerance. The joint angles generated by the simulation model were then compared with and applied to the physical manipulator. The results show that the MDH-based compensation model can align the bent wire tip with the insertion target and improve the feasibility of automated wire insertion using a closed commercial manipulator system.

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Chapter 1 Introduction

In the progression of modern industrial automation, although manipulators have demonstrated excellent repetitive positioning capabilities, however, all robotic systems cannot avoid geometric and non-geometric errors. These errors come from mechanical tolerance, joint backlash, structural deformations, even reach joint limit can have influence when gripper holding Flexible linear objects (FLOs) such as wires and cables which have physical deformability. Because gravity will make FLOs have uncertainty shape, this kind of task may include hand-eye coordination for real-time detection. Nevertheless, vision system may have original estimation error when sensing the random deformations, result in combining with manipulator's error, it would become complicated, that is the obstacle to achieving full automation in precision insertion task.

The motivation for focusing on integrated error analysis arises from two major industry bottlenecks. First, most commercial manipulator controllers operate as closed-loop "black-box" systems, which user can't get the internal kinematic algorithms making error compensation difficult. Second, although fields such as humanoid robotics and AI training are advancing by utilizing large-scale simulations to deal with complex deformations, the stability of AI models still based on a deep understanding of hardware-level errors. Physical and mathematical calculations cannot be entirely replaced by continuous trial-and-error model training; without precise physical modeling and defined boundaries, high-level AI control hard to maintain stability within extremely tight tolerance spaces.

This research uses a "Model-Oriented Integrated Analysis" to solve

these problems. The study first uses the Modified Denavit-Hartenberg (MDH) method to build a new model for the ABB IRB 1090 manipulator. This method helps the system move around the limits of closed software. It also creates a standard math base for the all six-axis manipulator.

Next, the research focuses on the angle of the wire. The study uses GeoGebra 3D and MATLAB to run simulations. This method fixes the wire's direction without changing the manipulator's tool center point(TCP). The wire aligns with the socket correctly. This process makes the precision insertion task much more successful.

1.1 Background

In the landscape of modern industrial automation, manipulators have been used in wide range of rigid assembly such as automotive industry and electronic industry, everything we used today may assembly by manipulators. However, as industrial requirements shift toward more complex assembly tasks, these task usually done by humans because the precise more higher and it takes lots of time to earn itself by automated, as the cost of building manipulators drops down, the automated handling of FLOs like wires, cables, and electronic soft materials has become a central technical challenge for automation field. Unlike rigid components, FLOs have stiffness and physical deformability making it easy to sagging and shape uncertainty shown in Fig. 1.1, causing traditional position-control strategies difficult to apply directly, changing a transition toward more accurate multi-sensor fusion and error-compensation mechanisms.

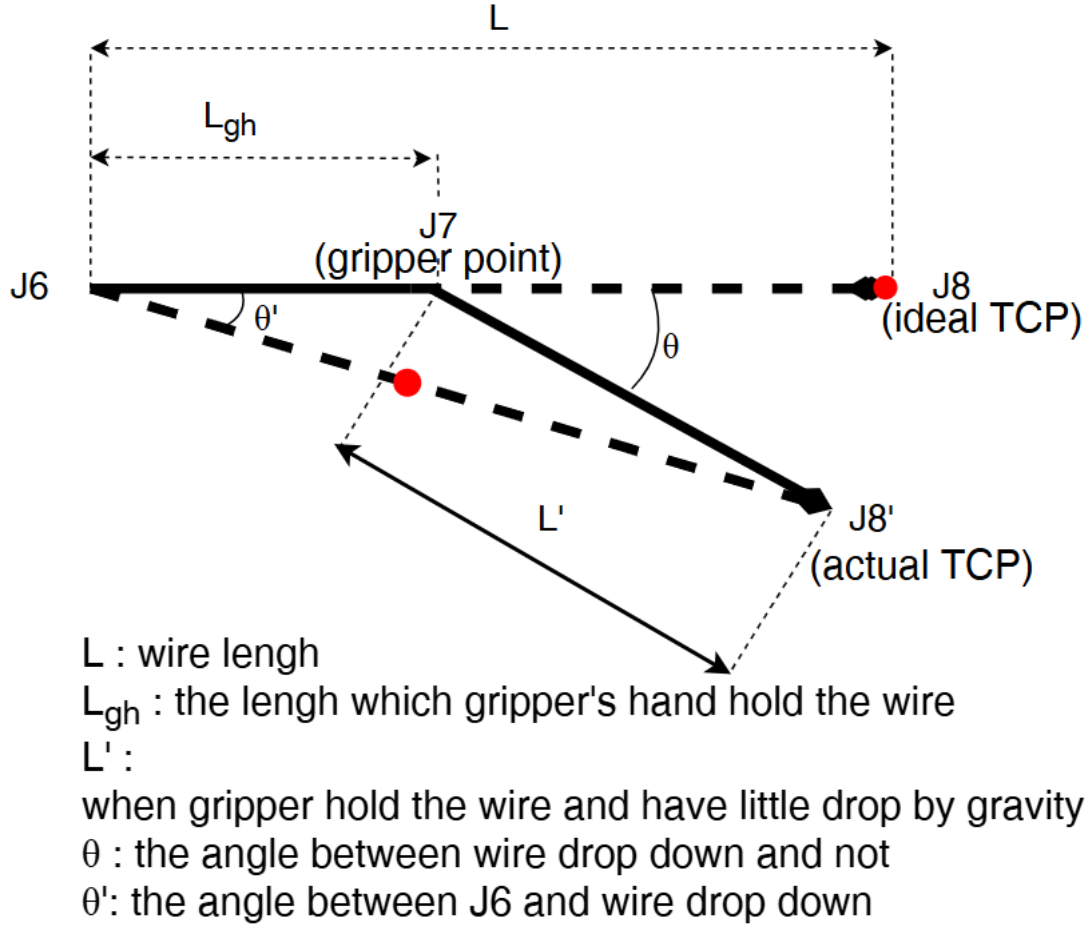


Figure 1.1: FLOs may deform by gravity

During this technical evolution, the positioning accuracy of robotic systems has faced unprecedented challenge. In precision insertion tasks, the tolerance clearance is often reduced to micro-levels (such as the 0.3mm aperture constraint discussed in this study), where even small deviations can lead to task failure. In practice, robotic systems possess a complex cumulative error chain, which includes geometric tolerances of the manipulator structure, pose offsets of end-effectors under varying design angles, and positioning uncertainties within the vision system during coordinate transformations.

Nowadays, the academic have a bottleneck because of the "closed" nature of commercial manipulator controllers in industry. The underlying kinematic algorithms of many brands are treated as "Black-box Systems" protected by intellectual property rights. Consequently, developers lack access to precise internal parameters and compensation logic, making it difficult to establish a universal mathematical criterion for quantifying system residuals during cross-platform error integration but using lots of extrinsic sensor to accomplish environment detection.

Furthermore, while Artificial Intelligence (AI) and Reinforcement Learning (RL) have made significant strides in humanoid robotics and complex deformation control by utilizing large-scale data training to address object uncertainty, the stability and generalization of AI models still have to grounded in a deep understanding of hardware-level errors from an engineering reliability perspective. Without precise geometric modeling as a computational baseline, it is hard to provide the precision assembly tasks only rely on trial-and-error training modes.

Therefore, establishing a "pose compensation model" that has product based and does not rely on specific controllers, through deeply exploring the error propagation logic among the manipulator body, the gripping tool, and the sensing system, and testing the model reliability, can provide a solid theoretical solution for precision automation systems.

1.2 Motivation

As electronic products and automotive components tend to be smaller and more precise, the difficulty of connector assembly tasks increases day by day. Currently, production lines still heavily rely on manual labor to plug and unplug wire harness terminals. However, long hours of high-precision work not only cause operator fatigue and lower the yield rate, but also face the challenge of a global labor shortage. Therefore, the automated handling of FLOs has become the core technology of the next stage.

Different from rigid objects, FLOs have physical characteristics of low stiffness and easy deformation. The physical sagging and shape uncertainty caused by gravity make traditional strategies based on "position control" difficult to apply directly, forcing a shift to rely on more precise multi-sensor fusion and error compensation mechanisms. However, in practice, robot systems have a complex cumulative error chain. Its sources include the geometric errors of the body structure, the bending condition of the wire during gripping, and the uncertainty of the vision detection system during the coordinate transformation process, making the positioning accuracy face a severe test in precision insertion tasks with micro-level tolerance.

Especially when using general-purpose industrial manipulators, developers often cannot access the internal closed kinematics algorithms. As a result, every time an object is gripped, the position and pose of the TCP are difficult to calibrate precisely, which further limits the customization capability of the robot parameters in complex tasks.

To overcome the above difficulties, this research proposes a "model-oriented compensation analysis method" as the core path. First, the study utilizes the Modified Denavit-Hartenberg (MDH) parameter method to remodel the geometric structure of the ABB IRB 1090 manipulator, thereby bypassing the limitations of closed algorithms and establishing a set of universal mathematical standards. Subsequently, this research focuses on the analysis of wire pose deviation, combining GeoGebra 3D calculator and MATLAB simulation to discuss how to correct the orientation of the wire through a pose compensation algorithm without changing the actual TCP, achieving the precise alignment of the terminal and the target socket, and significantly increasing the success rate of precision insertion tasks.

In this study, the TCP refers not only to the position of the wire tip but also to the orientation of the tool coordinate frame attached to it. Its formal position and orientation definitions are presented in Section 3.2.3.

1.3 Problem Statement

The handling of FLOs still faces significant challenges in the progression of precision assembly automation. First, FLOs possess physical characteristics of low stiffness and deformation easily, which cause uncertain sagging under gravity. This physical uncertainty makes it difficult for traditional "position control" strategies, based on rigid body geometry, to achieve stable insertion tasks directly.

Second, commercial manipulator controllers widely used in industry are mostly closed "black-box systems." Developers cannot access the in-

ternal kinematic algorithms and compensation logic. Therefore, it is difficult to accurately determine the distance and position of the fixture on the manipulator's end flange when the tool coordinates of the objects picked by the manipulator change every time. This limitation restricts the manipulator's ability to customize parameters for complex assembly tasks.

Finally, existing error analyses often treat the manipulator, the end-effector, and the vision sensors as independent entities. However, under a very small 1.5mm clearance constraint, a minor deviation from a single source will create a cumulative effect through coordinate transformations. Traditional position control strategies cannot correct this angular deviation. This posture mismatch will lead to geometric interference, mechanical jamming, or wire buckling during the insertion process. Therefore, without an integrated posture compensation method, the system will struggle to define stable successful boundaries for wire insertion, which eventually leads to an increased failure rate in automated assembly.

1.4 Research Gap

In the field of robotic assembly, although kinematic calibration techniques are highly mature, three critical research gaps still remain in the academic community:

1. The gap between specific objects and small tolerances: Existing studies mostly focus on the precision insertion of rigid objects. Although these studies reach high precision in their data, they use objects that do not rely on high precision for experimental testing. Systematic research

on "Flexible Linear Objects (FLOs)" with random deformation properties under a tiny 1.5mm range environment remains insufficient.

2. The disconnection between development environments and system closure: Many current studies on robotic arms utilize ROS (Robot Operating System), which allows researchers to design their own underlying control and kinematic algorithms. This development mode provides high transparency and customization space. However, commercial controllers widely used in industry are mostly closed "black-box systems," and developers cannot access the internal calculation logic. The academic community lacks a universal modeling method between open research systems and closed commercial systems.

3. Lack of an integrated error correction framework: In actual automated wire insertion systems, positioning and pose errors do not come from a single source. It includes the geometric tolerances of the manipulator, the manufacturing offset of the fixture and terminal gripping, and the uncertainty of hand-eye vision coordinate transformation. Previous literature mostly discusses "kinematic geometric calibration" and "vision-guided compensation" separately. When the tolerance of the wire product is small, these multi-source errors will generate non-linear coupling. Currently, the academic community lacks an integrated error correction framework that can combine multi-source errors into the same model without changing the absolute tool point position (TCP position), which prevents the yield rate of general-purpose manipulator in precision insertion tasks from further improving.

1.5 Research Objective

This study aims to solve the difficulties of flexible object handling and system error quantification in precision assembly. The specific objectives are as follows:

First, to validate the assembly feasibility of flexible linear objects under extremely small tolerances. This study will perform practical wire insertion experiments on FLOs with random deformation properties under a tiny 1.5mm clearance constraint. Through experimental data, this study will evaluate the specific impact of physical deformation on the assembly success rate.

Second, to develop a universal kinematics model applicable to closed commercial controllers. This research will utilize the MDH modeling method, paired with the MATLAB Robotics Toolbox, to establish a 6-axis manipulator simulation system that does not rely on the original manufacturer's underlying authority. This model will precisely define the geometric relationship between the fixture and the manipulator base, thereby bypassing the customization limitations of commercial "black-box systems."

Third, to implement and use the model to solve the situation where the wire bends when gripped by the manipulator. In the simulation environment, by compensating for the angle between the wire tip in space and the sixth axis of the manipulator, the end axis of the arm is aligned with the bent wire tip. The angles of each axis of the manipulator are calculated back and fed into the physical arm to move in front of the insertion socket surface. In this way, the physical manipulator position can be corrected using the

model under various wire bending angles to finally execute the insertion task successfully, and the description and transformation of the mathematical forms of the manipulator coordinate system will be discussed.

1.6 Scope of Study

The scope of this study focuses on the quantification and analysis of geometric errors in precision automated assembly. The boundaries are defined as follows: First, Hardware and Simulation Environment. This research only uses the 6-axis industrial manipulator as the experimental platform. It utilizes the MATLAB Robotics Tool box to establish a digital-twin model. All kinematic calculations and simulations are performed based on this specific hardware and software configuration.

Second, Research Target and Experimental Constraints. The research target is limited to Flexible Linear Objects (FLOs) fixed by the fixture. The experimental task is defined as inserting a wire into a fixed socket with a 1.5mm clearance. Additionally, the fixture geometry is restricted to the 4 directions 15-degree tilted fingers designed in this study.

Third, Analytical Methods and Technical Boundaries. This study emphasizes static kinematic modeling and the integrated analysis of multi-source error chains. The scope covers coordinate transformations of the manipulator and the fixture geometry. This research only provides a discussion of the vision system.

1.7 Significance of Research

This research is significant from both academic and industrial perspectives. For FLOs assembly, the proposed wire-insertion analysis and pose compensation method provide a model-based approach for evaluating and correcting wire-tip deviations before physical insertion. This can reduce the dependence on repeated manual adjustment and provides a feasible foundation for improving automation in high-precision wire assembly applications.

In addition, the MDH-based kinematic framework developed in this study operates externally from the commercial manipulator controller and does not require access to its internal kinematic algorithms. Therefore, the proposed approach can serve as a transferable modeling framework for different six-axis industrial manipulators by reconstructing their corresponding geometric parameters. Combined with the TCP pose compensation method, the framework also provides a basis for analyzing the influence of manipulator, fixture, and wire-related deviations under small-tolerance insertion conditions, and can be further extended to integrate manipulator calibration and vision-based feedback for more accurate automated assembly.

1.8 Organization of the Thesis

This introduction presents the research background and motivation, defines the problems faced in the precision assembly of flexible objects, and es-

establishes the research purposes, scope, and expected significance, which is followed by a literature review that covers the academic fields related to this study, including manipulator kinematic calibration, the automated handling of FLOs, and vision pose analysis methods.

The methodology and modeling part serves as the core theoretical foundation and experimental methods of this research, which covers using the MDH method to model the six-axis manipulator, and through the way of TCP displacement, recalculating a hypothetical TCP based on the actual TCP tilt angle and position using only the model, while the physical arm does not change the ideal TCP. Then, the experimental results and analysis integrate the target point, manipulator, and wire information through MATLAB for simulation, and the angles of each axis of the manipulator are calculated back through inverse kinematics, achieving that the physical arm can also precisely complete the wire insertion task without changing the physical TCP.

Finally, the conclusion and future work summarizes the key findings and technical contributions of this research, and provides suggestions for future research directions in vision compensation and full automation.

Chapter 2 Literature Review

This chapter reviews the literature related to robotic wire insertion and establishes the background for the proposed pose compensation method. Since wires are flexible, easily deformed, and affected by gravity, the actual wire tip may deviate from the ideal tool center point during manipulation. This deviation can reduce the success rate of precision insertion. Therefore, Section 2.1 first reviews studies on manipulator-based wire insertion and connector assembly to clarify the application background and technical motivation of this research.

Section 2.2 then discusses the relationship between wire types and AWG specifications, explaining how wire diameter and physical properties influence stiffness, bending behavior, and insertion tolerance. Section 2.3 reviews hand-eye coordination, vision-guided manipulation, and fully automated insertion procedures, focusing on how visual information supports manipulator positioning and alignment. Finally, Section 2.4 returns to manipulator kinematic modeling and discusses the use of the Modified Denavit-Hartenberg method to establish an analyzable six-axis manipulator model. Overall, this chapter develops the literature review from wire insertion applications to wire characteristics, vision-based collaboration, and kinematic modeling, forming the theoretical basis for the proposed pose compensation approach.

2.1 Manipulator-Based Wire Insertion

Before delving into fully automated cable and connector mating technologies, let us first review the current research landscape regarding robotic wire and connector assembly. In modern manufacturing, while the automated assembly of rigid objects is highly mature, tasks like wire harness assembly—which involve highly flexible Deformable Linear Objects (DLOs)—still face significant challenges. As Wang et al. [4] explicitly point out in their computer vision overview, the highly deformable and structurally unstable nature of cables means wire harness assembly still relies heavily on skilled manual labor. Achieving fully automated robotic cable or connector mating is not a simple pick-and-place task; rather, it requires a deep, systemic integration of visual perception, force feedback, gripper design, and mating strategies.

Regarding grasping strategies and pose estimation, the nearly infinite degrees of freedom of flexible cables make traditional modeling and control prohibitively expensive and complex. To address this, Zhou et al. [5] categorized wire harnesses with rigid connectors as Semi-Deformable Linear Objects (SDLOs), noting that directly and accurately estimating the spatial pose of the rigid connectors is the key to bypassing complex cable modeling and solving manipulation difficulties. However, even with visual assistance, stably grasping a connector directly from a cluttered pile remains a major hurdle. To overcome this, Song et al. [6] developed a dedicated cable connector-feeding system. Using an alignment mechanism, this system ensures the manipulator grasps the connector in a straight, secure manner every time, effectively replacing complex and time-consuming bin-picking

algorithms and laying a solid foundation for subsequent assembly. In real-world automotive production lines, Jiang et al. [7] significantly simplified the recognition task for the robotic vision system by attaching specially marked plastic shells to the wire harness clamps, enabling stable grasping and assembly through multi-arm collaboration.

In terms of hand-eye coordination and visual guidance, most studies employ a combination of global and eye-in-hand cameras to handle the uncertainties of the connector's initial pose. Zhou et al.'s system utilizes deep learning-based object detection for rough locating, followed by shape-based 3D matching for precise 6-DOF pose estimation, ensuring accurate positioning before grasping and mating. Furthermore, prior to executing the actual insertion, Di et al. [8] and Sun et al. [9] utilized CCD camera image matching to monitor and diagnose whether the connector was grasped correctly (e.g., detecting initial errors such as "not grasped" or "wrongly grasped"). This allows the system to pre-calculate initial deviation compensations, ensuring the connector perfectly aligns with the socket.

Finally, regarding force control and mating strategies, the extremely tight assembly tolerances of electronic connectors (for instance, a standard 8-pin Molex connector has a tolerance of only about 0.2 mm) dictate that visual positioning alone is insufficient for fully automated insertion. Song et al. proposed a tilt mating strategy based on impedance control. During the contact phase, they utilized a force/torque sensor combined with a micro up-down search motion to generate compliant contact, overcoming residual visual positioning errors and achieving a smooth insertion. Similarly, Di et al. and Sun et al. introduced force/torque sensors during the mating phase to detect contact patterns. If jamming occurs, the system automati-

cally triggers a blind search or torque control strategy for error recovery, ensuring the mating task is completed smoothly and reliably. Additionally, to address situations where the initial pose of the connector makes direct insertion difficult, Zhou et al. developed a dual-arm hand-changing strategy. Through a continuous sequence of grasping, reorienting, and releasing, the connector is adjusted into the optimal final pose for insertion.

To summarize, through the tight integration of visual tracking, force compliance, and grasp planning, manipulators have gradually developed mature solutions for the fully automated mating of flexible cable connectors. Following this, this research will further elaborate on your specific objectives, the different wire types, and the relevant MDH modeling for the manipulator.

2.2 Wire Type and Flexible Linear Object Deformation

To achieve fully automated wire harness assembly, this literature review first focuses on the ultimate goal of the process: robotic wire insertion and terminal mating. In automated assembly workflows, the core challenge of the insertion task lies in precisely aligning and inserting highly flexible, easily deformable wire ends into target sockets or connectors. For instance, in an automated plug task, a manipulator must overcome the spatial unpredictability of the wire, continuously tracking and correcting the cable-tip's posture to perfectly align its 6D pose with the target socket to ensure a successful insertion [10].

Before delving into robotic insertion, it is essential to clarify the wire's specifications and mechanical properties. American Wire Gauge (AWG) serves as the standard gauge system and acts as the underlying baseline specification affecting wire properties; it dictates the conductor's diameter, cross-sectional area, and the actual outer diameter of the wire—notably, stranded wires have a larger outer diameter than solid wires of the same AWG. However, AWG itself is not the primary subject of robotics control research. For robotic insertion, the true challenges arise from the physical phenomena dictated by these AWG specifications: wire diameter, terminal size, flexibility, bending capability, gravity sag, and the displacement of the wire tip after grasping.

In the robotics field, these challenges are substantially addressed by research on the mechanics and manipulation of Deformable Linear Objects (DLOs) or Flexible Linear Objects (FLOs) [11]. For example, wires exhibit significant sag and deformation under gravity. The bending behavior of both elastic and rigid wires can be analyzed using Euler's elastica model. This model treats the wire as an elastic, non-stretchable rod, enabling the calculation of its stable equilibrium shape when grasped and identifying the geometric conditions required to avoid self-intersection. Furthermore, parameters such as Young's modulus and shear modulus directly impact the stress feedback and deformation states the manipulator experiences during manipulation [12].

To overcome the severe deformation challenges posed by DLOs/FLOs, hand-eye coordination and visual feedback are indispensable for fully automated insertion [13]. This progression typically begins with environment and target perception, where a global camera first provides an overview of

the workspace to roughly locate connectors or sockets. A hand-eye camera mounted on the manipulator’s end-effector then maneuvers over the target to perform shape matching and high-precision 6D pose estimation for fine positioning [14].

Concurrent with target perception, the system executes wire modeling and pose tracking prior to grasping. By filtering noise from point cloud data and applying mathematical models—such as quadratic polynomials—to parameterize the flexible wire’s 3D geometry, the vision system enables real-time estimation of the ”cable-tip” pose and calculates an optimal grasp point at an appropriate distance from the tip.

As the operation advances to the PRE-INSERT stage, the manipulator activates a pose alignment controller (e.g., a PD controller) for dynamic correction. Relying on real-time visual feedback of the tip’s pose deviation, this controller calculates the requisite Cartesian translational and rotational velocities for the end-effector to dynamically eliminate alignment errors between the tip and the socket. To safely navigate complex environments, this control loop integrates a Global Planner with a Local Controller, utilizing Control Barrier Functions (CBFs) to ensure the wire does not experience damaging overstress or collide with environmental obstacles during bending and insertion.

Ultimately, bridging these visual and control frameworks requires precise MDH modeling; only through accurate kinematic mapping can the wire-tip features and socket coordinates captured by the hand-eye camera be correctly translated to the manipulator’s Base Frame, endowing the system with the capability to overcome wire deformation and execute high-

precision, fully automated insertion.

2.3 Hand-eye Coordination, Vision Guidance, and Fully Automated Insertion

In the field of automated assembly, wire harness and connector insertion tasks are characterized by flexibility and deformability. Coupled with a wide variety of dimensions (e.g., AWG specifications) and exterior designs, traditional robotic control relying on fixed trajectories is no longer applicable [15]. Furthermore, the superimposition of measurement errors from the vision system and kinematic errors from the manipulator itself poses a significant obstacle to fully automated, high-precision insertion. Consequently, this process must rely on real-time hand-eye coordination and multi-sensor feedback correction [16]. In this operational workflow, the first step in visual guidance is capturing environmental and target information. Due to the complex structures and non-rigid nature of connectors and wires, early image processing techniques heavily relied on external markers or carefully engineered backgrounds [17]. To overcome this limitation, recent studies have adopted deep learning architectures (e.g., YOLOv5, Faster R-CNN, and Mask R-CNN) that empower systems to directly recognize various connector specifications and extract the 3D geometric profiles of thin wires within complex environments. To counteract geometric extraction errors caused by wire deformation, researchers have also proposed smoothing algorithms based on depth values to provide more accurate target feature information.

To translate the target identified in the image into executable robotic actions, the system must then perform precise hand-eye calibration. Regardless of the camera's mounting configuration [18], the goal is to calculate the homogeneous transformation matrix between the camera coordinate system and the manipulator's end-effector (or base) coordinate system [19,20]. Classic algorithms have traditionally utilized dual quaternions, global polynomial optimization, or nonlinear least squares methods to simultaneously solve for the rotation and translation matrices, while cutting-edge techniques further integrate 3D point clouds with deep learning to achieve marker-less automatic calibration by scanning the manipulator base [21]. Once calibration is complete, the connector or target slot pose estimated by the vision system can be seamlessly transformed into the manipulator base coordinate system.

Based on this information, the manipulator controller calculates the optimal grasping position for the wire and plans an "approach pose" to move toward the target insertion hole. However, if the system relies solely on an open-loop "look-and-move" architecture, this stage is highly susceptible to positioning offsets due to the accumulated errors from visual measurements and manipulator kinematics. Ultimately, through the dual verification of vision and force, the system ensures that the terminals are correctly seated while overcoming insertion and extraction resistance. If the insertion process encounters an obstruction, impedance control allows the manipulator to compliantly retract and realign, establishing a comprehensive error recovery mechanism that guarantees the final assembly quality.

2.4 MDH-based Manipulator Modeling and Kinematic Calibration

In the fully automated wire insertion tasks discussed previously, whether addressing the deformation characteristics of different AWG wires or relying on hand-eye systems for precise collaboration, success hinges on the robotic manipulator possessing exceptionally high absolute positioning accuracy. Achieving this accuracy, however, requires a fundamental examination of the manipulator's underlying geometric modeling and error compensation.

The kinematic foundation of robotic manipulators traces back to the DH convention, which utilizes matrix algebra to establish a complete symbolic description and mathematical equations for lower-pair mechanisms. The primary value of DH parameters lies in their ability to clearly define the relative link-to-link displacements using homogeneous transformation matrices, rendering the spatial state of the mechanism both repeatable and derivable [22].

However, the traditional DH model frequently encounters ambiguities in coordinate frame definitions when applied to links with more than two joints, tree structures, or closed-loop systems. To resolve these issues, researchers introduced the MDH convention, which fixes the coordinate frames directly to the joints of the links, thereby eliminating the ambiguities of the traditional DH method and providing a more intuitive kinematic definition suitable for both open- and closed-loop robots. By utilizing MDH parameters (comprising joint angle θ , link offset d , link length a , and twist

angle α), the spatial geometric relationship from the base to the end-effector can be precisely calculated [23].

Although the MDH theory provides an elegant kinematic framework, relying solely on the manufacturer’s nominal MDH parameters is highly insufficient in real-world automated insertion scenarios. Literature indicates that while actual industrial manipulators exhibit excellent repeatability, their absolute accuracy is often inadequate due to a combination of geometric and non-geometric errors. Geometric errors, arising from manufacturing and assembly tolerances, cause inherent parametric variations—such as joint offsets and link length inaccuracies—between the real manipulator and its nominal model. Concurrently, non-geometric errors further degrade this absolute accuracy through nonlinear factors, including compliance (or stiffness) in gearboxes and links, backlash, joint friction, and gravity-induced sag. For instance, gravity and changes in payload can introduce significant tangential errors at the joints, with elastic deformations amplifying as the distance from the axis of rotation increases.

To mitigate these errors, the existing literature extensively employs manipulator kinematic calibration techniques. The calibration process generally consists of three steps: modeling, measurement, and parameter identification. During the parameter identification phase, researchers typically formulate an objective function based on the discrepancy between the measured actual end-effector poses and those predicted by the nominal model; nonlinear optimization algorithms (such as least squares or quasi-Newton methods) are then applied to determine the MDH compensation parameters

$$(\Delta\theta, \Delta d, \Delta a, \Delta\alpha)$$

that best reflect the true physical behavior [24]. Advanced calibration studies even introduce non-geometric parameters—such as specific compliance coefficients designed for individual joints—alongside fundamental geometric parameters to model the dynamic elastic deformation of the manipulator [25]. Whether utilizing high-precision laser trackers or nonlinear optimization methods based on monocular vision and cameras, the core objective remains the same: to identify the deviations between the actual manipulator and the theoretical model, and to compensate for these parameters accordingly.

In summary, both the theoretical foundations of MDH and the extensive calibration literature confirm that to perfectly execute high-precision tasks like wire insertion, the inherent geometric and non-geometric errors of the manipulator must be addressed and compensated for. This evidence perfectly reinforces the rationale for the "model-oriented compensation" framework established in this study.

Since the underlying control algorithms of most contemporary commercial manipulator controllers are closed-system "black boxes," the objective of this research is not to supersede these commercial controllers. Rather, it aims to develop an external, mathematically analyzable geometric compensation model based on the MDH convention. Through this add-on model compensation architecture, we can inversely transform the end-effector deviations observed by the hand-eye collaboration system into precise kinematic compensations. This approach effectively eliminates spatial errors introduced during link manufacturing and assembly, thereby realizing a fully automated and highly adaptable wire insertion system.

2.5 Review of Key Studies

The reviewed studies show that fully automated wire insertion is a complex task involving flexible-object manipulation, visual perception, force-guided assembly, and manipulator kinematic compensation. Unlike rigid parts, wires and cables behave as DLOs or FLOs, whose shapes vary under gravity, grasping force, and environmental constraints. Therefore, successful robotic insertion requires not only accurate manipulator positioning, but also stable grasping, connector pose estimation, and reliable mating strategies.

Previous studies on wire harness and connector assembly have addressed these challenges through different system designs. Jiang et al. demonstrated wire harness assembly using multi-arm collaboration, visual tracking, and specially designed markers or fixtures to simplify recognition and manipulation. Zhou et al. introduced the concept of SDLOs, emphasizing that estimating the rigid connector pose can reduce the difficulty of modeling the entire flexible cable. Other studies further combined connector-feeding mechanisms, force/torque sensing, impedance control, and fault-tolerant recovery strategies to handle tight tolerances and contact uncertainties during connector mating.

Wire specifications also play an important role in insertion performance. Although AWG is mainly a wire-size standard, it affects conductor diameter, outer diameter, stiffness, bending behavior, terminal size, and insertion tolerance. These physical properties determine how much the wire tip may deviate from its expected position after grasping. Thus, robotic wire insertion cannot be treated simply as a rigid tool-positioning problem.

Hand-eye coordination and vision-guided control provide the perception basis for full automation. Cameras can identify connectors, sockets, wire tips, and target poses, while hand-eye calibration transforms visual information into the manipulator base frame for grasp planning and alignment. However, vision-only positioning is often insufficient because visual errors and manipulator kinematic errors accumulate during execution.

Finally, DH/MDH modeling and calibration studies explain the need for a model-oriented compensation framework. Although MDH provides a systematic way to describe manipulator kinematics, real industrial manipulators still suffer from geometric and non-geometric errors. Therefore, this study builds on prior work by developing an external MDH-based pose compensation model to correct the deviation between the ideal TCP and the actual bent wire tip, improving the feasibility of high-precision automated wire insertion.

Chapter 3 Methodology

3.1 Methodological Framework

To clarify the logical and analytical workflow of this study. As shown in Figure 3.1, the research is divided into five phases.

The first phase involves manual wire insertion to confirm that the insertion action can be executed, the fixture and gripper designs have no issues but also gripper force is enough, the manipulator works within a normal workspace, and there are no singularity problems. The position of the insertion target and TCP are also recorded during this phase.

The second phase extracting the parameter which call MDH from manipulator to building the model on MATLAB. Based on MDH modeling principles and referencing the rotation directions and orientations of each axis of the physical manipulator, a universal model can be established in MATLAB. This model operates independently of commercial controllers and serves the subsequent simulations.

The third phase develop the equation of compensation to TCP position fixing. Although the ideal TCP and actual TCP at different locations, this study keeps the ideal TCP unchanged on the physical manipulator. Instead, the model recalculates an imaginary TCP based on the tilt angle and position of the actual TCP. This approach attempts to correct the bending angle within the simulation environment, and a GeoGebra 3D calculator is used to demonstrate the calculation method for the TCP correction, proving the

mathematical derivation is correct.

The fourth phase combines the first through third phases to establish a simulation environment in MATLAB that closely matches the physical environment, displaying the target point, manipulator, and TCP. Using the MATLAB toolbox, the actual TCP of the bent wire is rotated according to the derivation method from the GeoGebra calculator. This allows the imaginary TCP to align with the insertion socket. After using inverse kinematics to calculate back the angles of each axis in this aligned state, the setup is reproduced in the physical environment to confirm that the calculation method of the simulation environment is correct.

The fifth phase recall how the practical exercise succeed and the method used in this study. Last, discussing the application areas of this research and the goals that can be achieved in the future.

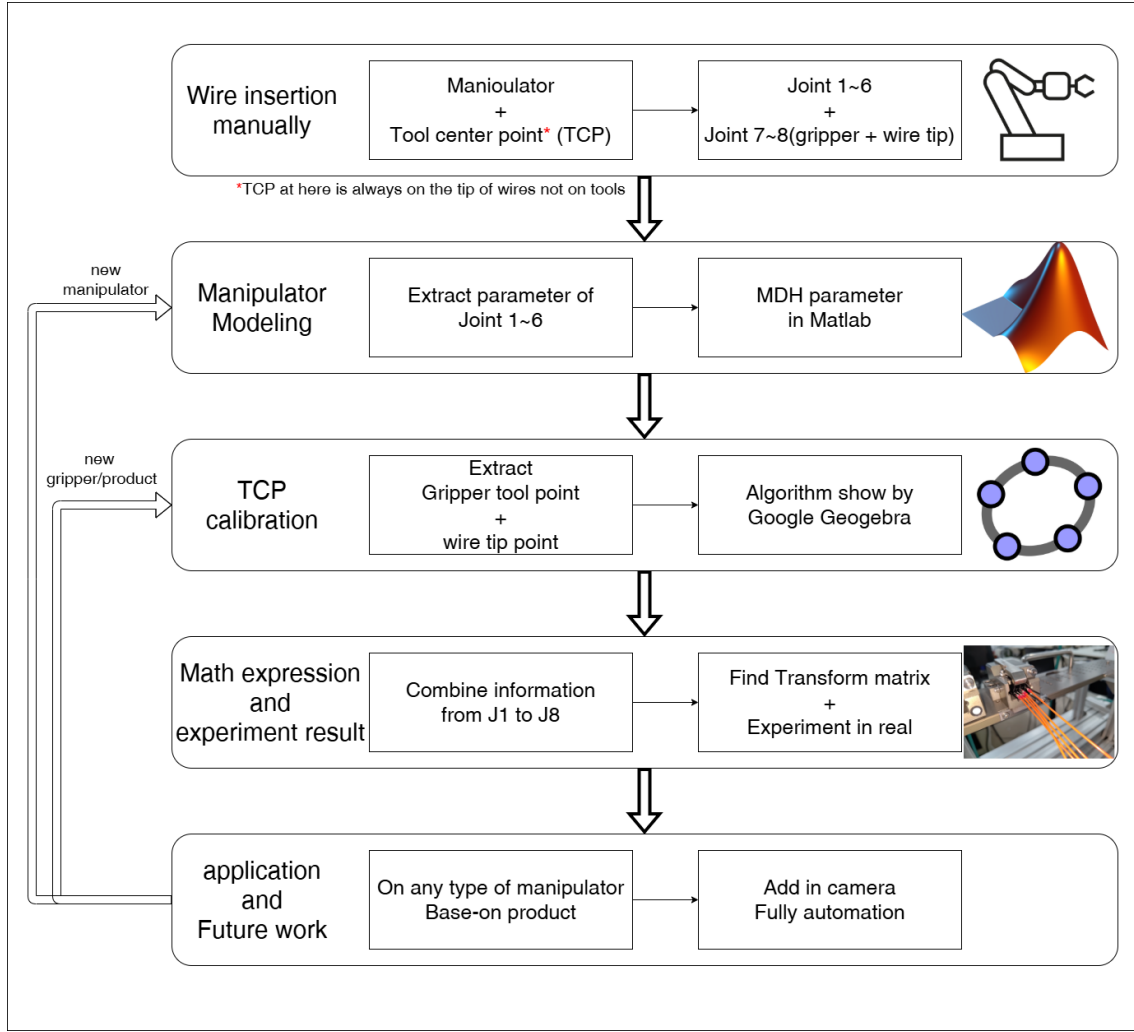


Figure 3.1: Schematic diagram in this research

3.2 Kinematic Modeling

To bypass the limitations of commercial controllers, this study models the manipulator using a fundamental approach. It can simulate the subsequent insertion system, and considering that the underlying kinematics logic of commercial manipulator is usually hidden, thereby allowing the method proposed in this research to be applied across different commercial manipulators.

3.2.1 Transformation Matrix Derivation

This subsection aims to establish the forward kinematics model of the robotic manipulator. Through the derivation of homogeneous transformation matrices, the spatial position and pose relationships between the manipulator links can be systematically described. Based on the previously defined DH parameters, here will step-by-step establish the mathematical mapping between adjacent coordinate frames, ultimately constructing the complete kinematics equation that describes the end-effector relative to the base coordinate frame. This serves as the theoretical baseline for subsequent error compensation and path planning. In kinematics, assuming a coordinate frame is used to describe a point^A P ,

$${}^A P = \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix} \quad (3.1)$$

However, kinematics frequently requires one coordinate frame to describe a point located in another coordinate frame, which represents a relative coordinate definition. If there is another coordinate frame B in space, this frame can be decomposed into unit vectors along three directions: $\hat{X}_B$, $\hat{Y}_B$, and $\hat{Z}_B$. When coordinate frame A is used to describe coordinate frame B, it is expressed as ${}^A_B R$. This represents the concept of a rotation matrix, indicating the orientation of frame B relative to frame A. Therefore, it can be written as:

$${}^A_B R = \begin{bmatrix} {}^A \hat{X}_B & {}^A \hat{Y}_B & {}^A \hat{Z}_B \end{bmatrix} \quad (3.2)$$

When coordinate frame B is described using coordinate frame A as the reference frame, it can be defined as:

$$\{B\} = \{{}_B^A R, {}^A P_{BORG}\} \quad (3.3)$$

Here, ${}^A P_{BORG}$ represents the position of the origin of coordinate frame B relative to the reference coordinate frame A. When a vector ${}^B P$ needs to be expressed in coordinate frame A as ${}^A P$, it is equivalent to transforming ${}^B P$ into coordinate frame A and adding the representation of the origin of frame B in frame A. The transformation between different coordinate frames applies the rotation matrix:

$${}^A P = {}_B^A R {}^B P + {}^A P_{BORG} \quad (3.4)$$

${}_B^A R$ describes the representation of ${}^B P$ under coordinate frame A, showing how ${}^B P$ looks in frame A after the transformation by ${}_B^A R$. Meanwhile, ${}^A P_{BORG}$ is the representation of the origin of coordinate frame B under frame A. This formula expresses the basic transformation mapping of a vector from one frame to a second frame through a standard vector rotation followed by a translation. To make the mathematics more concise, the concepts of rotation and translation are combined together into a single matrix:

$${}^A P = {}_B^A T {}^B P \quad (3.5)$$

${}_B^A T$ represents the transformation matrix, acting as an operator that directly expresses all elements related to coordinate frame B in terms of coordinate frame A. Fully expressed in matrix form, it is written as:

$$\begin{bmatrix} {}^A P \\ 1 \end{bmatrix} = \left[\begin{array}{ccc|c} {}_B^A R & & & {}^A P_{BORG} \\ 0 & 0 & 0 & 1 \end{array} \right] \begin{bmatrix} {}^B P \\ 1 \end{bmatrix} \quad (3.6)$$

This 4×4 matrix is also called a homogeneous transformation matrix. Simply put, it is a format used to describe both rotation and translation simultaneously. The concept of the homogeneous transformation matrix is identical in manipulator kinematics. The six-axis robotic manipulator used in this study has, as the name implies, six axes starting from the base. After transforming through six consecutive transformation matrices, the position and orientation of the sixth axis relative to the base coordinate frame can be obtained and written as:

$${}^1_6T = {}^1_2T * {}^2_3T * {}^3_4T * {}^4_5T * {}^5_6T \quad (3.7)$$

3.2.2 Coordinate Frame Assignment

To fully simulate the physical manipulator, the first step before modeling is to record the forward and reverse rotation of each link. Taking the physical manipulator in this study as an example, as shown in the figure 3.2, the forward and reverse rotation directions can be determined from the product manual of the commercial manipulator. This serves as the baseline for verifying whether the modeled manipulator aligns with the physical one.

Since a six-axis manipulator consists of six links, it features six coordinate frames, each corresponding to an individual axis.

When modeling the manipulator, the base coordinate frame is the most critical part and is considered the 0th axis. This coordinate frame remains stationary, but its orientation is identical to that of the first axis. If the orientation of the base coordinate frame is set incorrectly, all subsequent coordinate frames will be affected.

Generally, when modeling a manipulator base coordinate frame, the z_1 is aligned with the center of the joint rotation axis, and its direction is determined by the right-hand rule to match the rotation direction specified in the product manual. As shown in the figure, when the forward rotation direction of the manipulator base (i.e., the first axis) corresponds to the four fingers curling while the thumb points upward, it can be confirmed that the z_1 -axis of the base coordinate frame points upward. Subsequently, the directions of x_1 and y_1 can be aligned based on the linear motion function of the physical manipulator or can adhere to the world coordinate system within the controller.

Starting from the second axis, the definition of the coordinate frames becomes slightly more flexible. The direction of z_2 is determined in the same manner as z_1 , focusing on the center of the rotation axis and applying the right-hand rule. The directions of x_2 and y_2 can be decided by the user, but they must maintain the three-dimensional orientation tracking back to the base coordinate frame, allowing rotation while the z axis direction is fixed. To facilitate the subsequent establishment of MDH parameters, completely opposing directions to the base coordinate frame are typically avoided.

For the third axis, the direction of z_3 is determined using the same method as before. Since the third axis rotates in the same manner as the second axis, identical coordinate frame orientations are established here.

Starting from the fourth axis, the rotation axis direction changes, yet the method for determining the z_4 direction remains unchanged. The directions of x_4 and y_4 are chosen based on user convenience; here, they are set to

align with the direction of x_3 . The y_4 direction is then determined based on the established z_4 and x_4 directions. Note that the orientations across all three dimensions result from the rotation of the base coordinate frame.

The rotation axis of the fifth axis is perpendicular to that of the fourth axis; hence, the z_5 direction is perpendicular to z_4 . The direction of x_5 is set to match the preceding ones, which then determines the y_5 direction.

The sixth axis represents the end of the manipulator. The z_6 direction is perpendicular to the flange face and points toward the front of the manipulator. The x_6 axis is configured to point toward the left side of the manipulator, thereby determining the y_6 direction.

3.2.3 MDH Parameter and Definition

In the field of robotic manipulators, the DH method is most commonly used to establish the relationships between the axes of the manipulator. The DH method describes the distances and angular offsets between adjacent axes while strictly adhering to the XYZ orientations of each coordinate frame. Because different XYZ orientations inevitably lead to distinct final arm postures, a clear understanding of the purposes of the four DH parameters is essential for building a six-axis manipulator model.

The four parameters are listed in the table. It is worth noting that a modified DH (MDH) method is employed here, and the differences will be explained in detail in the next section.

These four parameters define the relationship between two adjacent

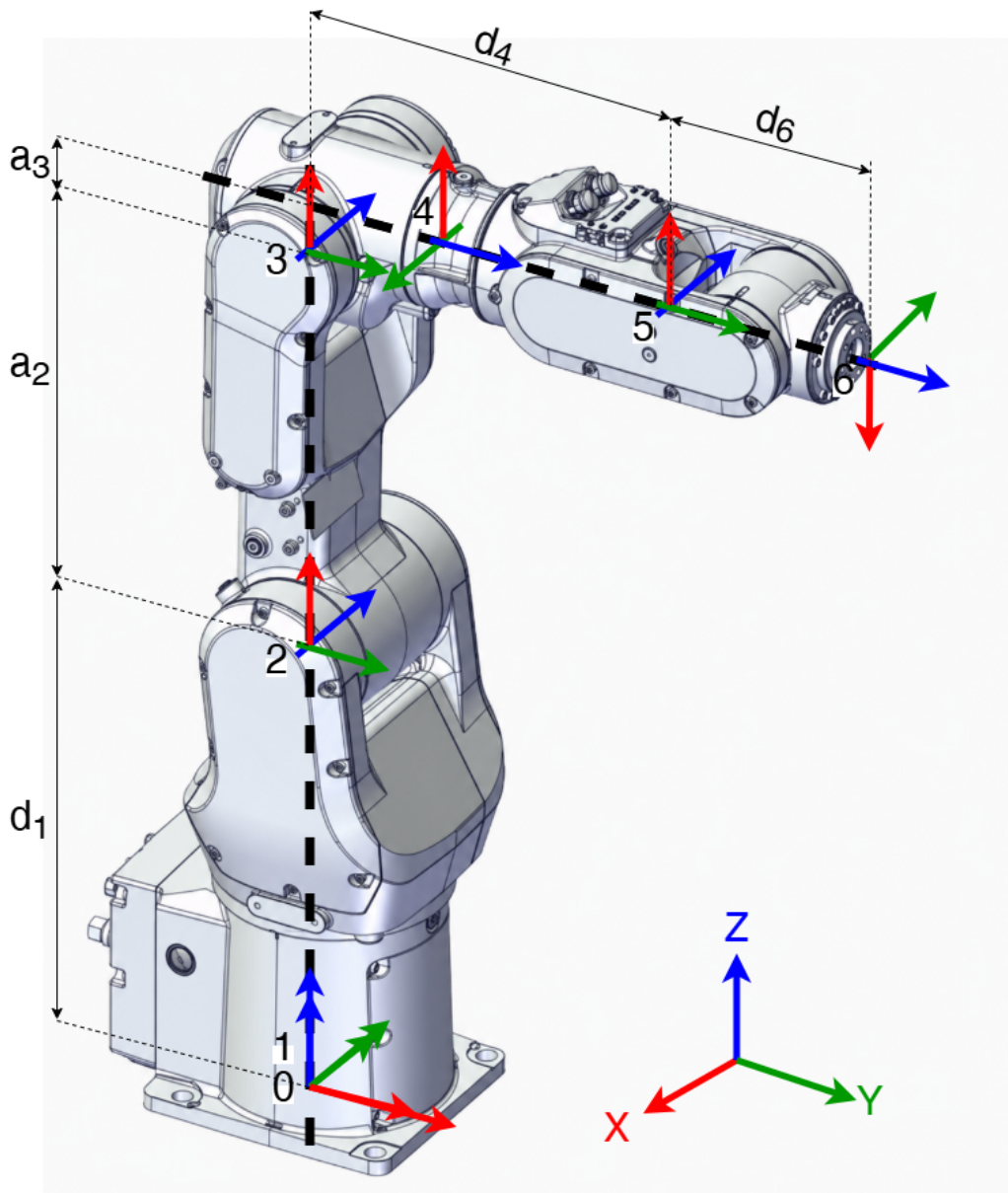


Figure 3.2: Direction coordinate frame for each joint [1]

axes. In the MDH convention, the coordinate frame $\{i\}$ is attached to the centerline of joint i as shown in the figure 3.3. The four parameters for the transformation from coordinate frame $\{i-1\}$ to $\{i\}$ are defined as follows:

- **Link length** a_{i-1} : The distance moved from Z_{i-1} to Z_i along the X_{i-1} axis. Measurement relationship: Describes the length of the

common normal between the two Z axes.

- **Link twist** α_{i-1} : The angle of rotation from Z_{i-1} to Z_i about the X_{i-1} axis. Measurement relationship: Describes the relative angle between the two Z axes, where the sign of the angle is determined by the clockwise or counterclockwise direction of rotation from the Z_{i-1} axis to the Z axis.
- **Link offset** d_i : The distance moved from X_{i-1} to X_i along the Z_i axis. Measurement relationship: Describes the length of the common normal between the two X axes.
- **Joint angle** θ_i : The angle of rotation from X_{i-1} to X_i about the Z_i axis. Measurement relationship: Describes the relative angle between the two X axes, where the sign of the angle is determined by the clockwise or counterclockwise direction of rotation from the X_i axis to the X axis.

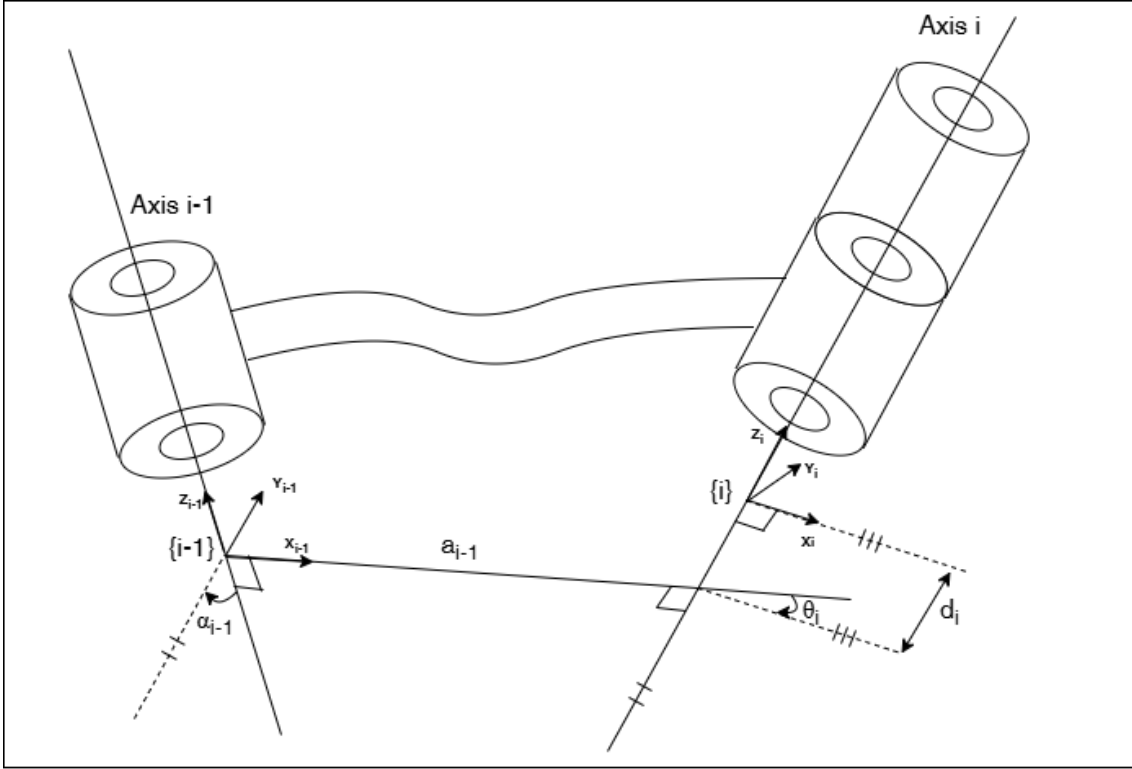


Figure 3.3: The relationship between two joint and can be described with four parameter [2]

After determining the coordinate frame orientations for each link, the geometric dimensions of the manipulator are required to establish the complete MDH parameter table. These dimensions can be obtained from the product manual of the manipulator; however, the main difficulty lies in determining whether each geometric distance should be represented by the link length a_{i-1} or the link offset d_i . According to the MDH convention, a_{i-1} represents the distance measured along x_{i-1} between two consecutive joint axes, whereas d_i represents the distance measured along z_i . Therefore, the corresponding parameter must be determined according to the geometric relationship and direction between adjacent coordinate frames.

For the first geometric dimension, since z_1 and z_2 intersect and no com-

mon normal exists between them, the corresponding link length is zero, while the vertical displacement is represented by d_1 according to the dimensions given in Table 3.1. Starting from the second dimension, the directional relationships between adjacent coordinate frames must be considered. Between the second and third joint axes, the separation is measured along x_2 from z_2 to z_3 ; therefore, this dimension is represented by a_2 . Similarly, the third dimension corresponds to the distance from z_3 to z_4 measured along x_3 , and is therefore denoted as a_3 . The following geometric displacement associated with the fourth joint is represented by d_4 . No additional translational parameter is required between the fourth and fifth joints because this relationship is represented only by a rotational transformation in the adopted coordinate-frame assignment. Finally, the distance associated with the sixth joint is represented by d_6 according to the direction of the corresponding z -axis. It should also be noted that the final link does not require an additional a_6 because there is no subsequent joint axis from which another common-normal distance must be defined. In contrast, offset parameters d_i may exist from d_1 to d_6 depending on the adopted coordinate-frame assignment.

After the translational parameters are determined, the remaining MDH parameters are the link twist α_{i-1} and the joint angle θ_i . The link twist describes the fixed angular relationship between two consecutive joint axes, whereas the joint angle represents the rotational state of each revolute joint. Therefore, α_{i-1} is determined from the geometric coordinate-frame assignment, while θ_i varies according to the manipulator configuration. For example, α_1 is determined by observing the rotation from z_1 to z_2 about x_1 . According to the right-hand rule, with the thumb pointing along the posi-

tive direction of x_1 , the curling direction of the fingers defines the positive rotation direction. Based on the coordinate-frame assignment used in this study, the rotation from z_1 to z_2 is 90° , resulting in $\alpha_1 = 90^\circ$. The remaining link twist angles are determined using the same procedure, completing the MDH parameter configuration summarized in Table 3.1.

Table 3.1: DH Parameters Table

Frame	α_{i-1}	a_{i-1}	d_i	θ_i
1	0°	0	327(d1)	0°
2	-90°	0	0	-90°
3	0°	280(a2)	0	0°
4	-90°	10(a3)	300(d4)	0°
5	$+90^\circ$	0	0	0°
6	-90°	0	64(d6)	180°

3.2.4 Comparison between SDH and MDH

In industrial manipulator kinematic analysis, researchers widely apply homogeneous transformation matrices to describe the position, orientation, and transformations between adjacent coordinate frames of a rigid body. The two most common and universal parametric modeling methods are the classical Standard Denavit-Hartenberg (SDH) model and the Modified Denavit-Hartenberg (MDH) model. Although both methods utilize four link parameters (link length, link twist, link offset, and joint angle) and 4×4 transformation matrices to establish kinematic equations through recursive iteration, they possess fundamental differences in coordinate frame

attachment and definition rules [26].

The core geometric difference between the two models lies in the alignment baseline of the Z-axis relative to the joint axis. By definition, the Standard DH model aligns the Z-axis of the previous frame (Z_{i-1}) with the current joint axis i . In contrast, the Modified DH model aligns the Z-axis of the current frame (Z_i) directly with joint axis i . Furthermore, the X-axis (X_i) of the MDH model points along the common normal from joint i to joint $i + 1$.

These different rules directly lead to distinct mathematical structures in the resulting homogeneous transformation matrices for SDH and MDH. In practice, researchers often face difficulties when choosing a modeling method for specific robots. Switching between algorithms or re-modeling due to different conventions results in a significant loss of time. However, although their definitions differ, these two models can map to each other and convert rapidly through specific mathematical relationships. Therefore, clarifying the geometric differences and transformation relationships between SDH and MDH models has significant practical importance for the subsequent development and optimization of manipulator kinematics, dynamics, and control algorithms.

In this study, emphasizing the relationships between adjacent axes, we primarily use the parameters described by MDH convention to perform the modeling of the manipulator.

3.2.5 MDH parameter and modeling in MATLAB

In the verification stage of the MDH parameter table, this study utilizes MATLAB to simulate the overall state of the manipulator. As shown in the figure 3.4, by employing the MATLAB Robotics Toolbox, it is possible to directly model and animate the manipulator. Sliders can be used to verify the rotation directions of the arm and determine whether the modeling is correct. If there are any errors in the established MDH parameter table, discrepancies between the simulated rotation directions and the physical ones will become highly apparent at this stage. In such cases, it is necessary to re-evaluate where the definition errors occurred in the MDH parameter table. This study extracts the MDH parameter table relying solely on the link lengths from the product manual and the rotation directions of each axis of the physical manipulator. The parameters are then implemented via code into MATLAB for simulation, establishing the baseline for the subsequent TCP correction algorithm.

3.3 Tool Center Point Offset Calculation

In this chapter, this study investigates the practical challenges encountered during manipulator insertion tasks and provides comprehensive solutions and calculation workflows through a mathematical model.

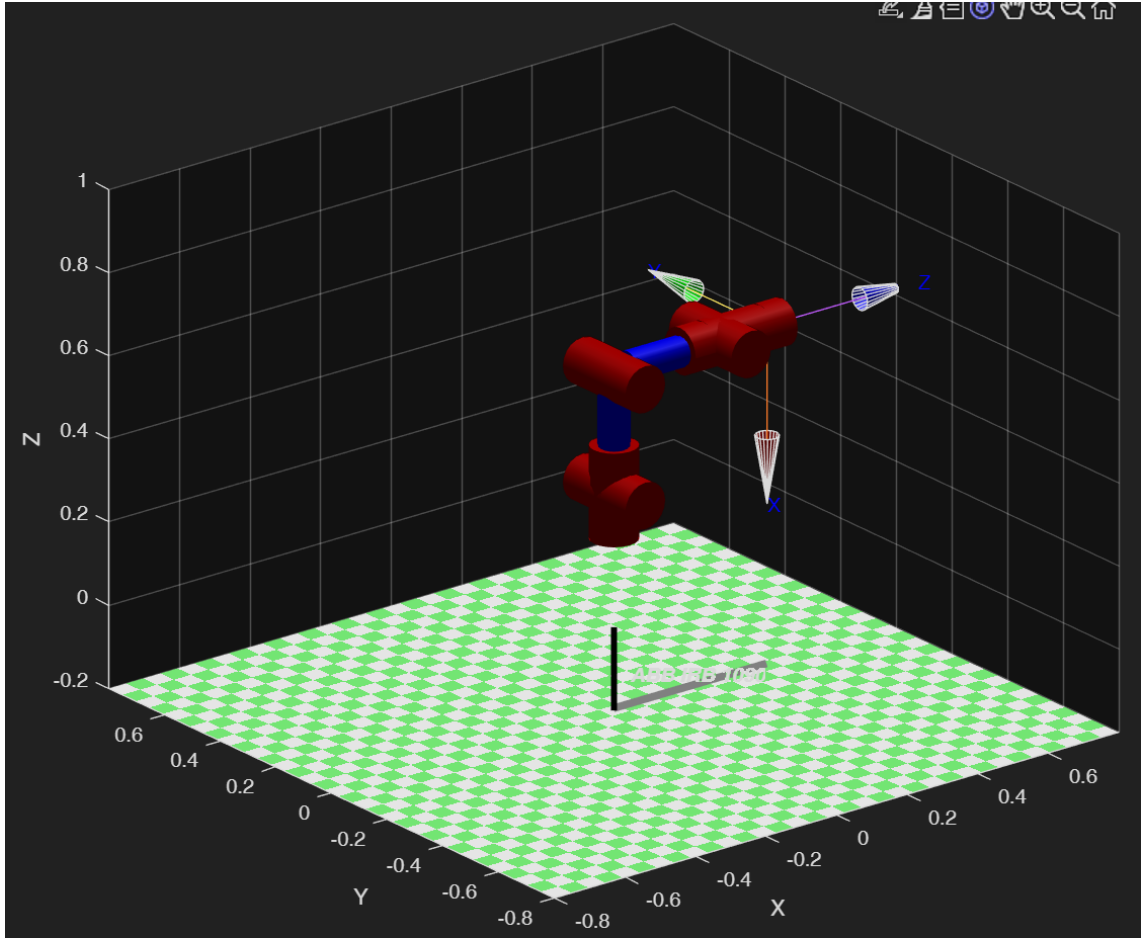


Figure 3.4: Simulate manipulator by MDH parameter [3]

3.3.1 Wire Insertion Problems

Various solutions are already widely available in industry for different robotic insertion tasks; however, most of these applications involve relatively low precision requirements. For instance, object placement can be broadly regarded as a type of insertion task. In this study, the experiments focus on slender and directional DuPont wires and their corresponding insertion sockets. This configuration requires the wire tip to approach the socket with an appropriate position and orientation in order to achieve successful insertion.

Through repeated physical experiments, it was observed that although the wire fixture provides a consistent initial condition and facilitates wire grasping, friction between the wire and the fixture during the grasping process may still cause the wire to bend. Once such deformation occurs, even if the position of the calibrated TCP is correctly defined in the manipulator controller, the actual wire-tip position and orientation may deviate from the ideal condition. For a high-precision insertion task, a visible angular deviation of the terminal wire segment can therefore significantly increase the likelihood of insertion failure. Accordingly, this section investigates how the wire-tip angular deviation can be compensated through the proposed mathematical model without physically redefining the calibrated TCP stored in the manipulator controller, thereby aligning the terminal wire segment with the target socket.

It should be noted that the accuracy of the physical insertion process is also affected by several other factors, including the positioning accuracy of the manipulator, fixture errors, TCP calibration errors, and manufacturing tolerances of the wire and socket. However, compared with the visible angular deformation of the wire observed in this study, these individual manufacturing tolerances are not treated as the primary source of pose deviation. Therefore, this study does not separately quantify each physical manufacturing error, but instead focuses on the modeling and compensation of the wire-tip angular deviation, focusing instead on analyzing and solving the angular issue.

3.3.2 Ideal, Actual, and imaginary Compensated TCP

Generally, the TCP is typically configured at the position of the gripper fingers, referred to as the gripper tool point. This setup is common because the objects gripped by the arm in most tasks feature high rigidity and theoretically act merely as an extension of the end-effector. However, to accurately determine the position of the wire tip, this study configures the TCP at the wire tip instead of the gripper fingers, defining it as the wire tip point. When the actual TCP on the end-effector tilts relative to the ideal TCP, the physical manipulator keeps the ideal TCP unchanged, utilizing the model to recalculate an imaginary TCP based on the tilt angle and position of the actual TCP. It can also be described as the wire tip (actual TCP) being initially misaligned with the socket, and the imaginary TCP is a new TCP configured specifically to resolve this alignment problem. The imaginary TCP varies according to the actual TCP—meaning its final position changes based on the wire tip position—with the sole purpose of aligning with the wire socket surface to ensure that the orientation of the wire tip allows for normal insertion into the wire socket surface.

3.3.3 Explanation Of Algorithms By Geogebra

Figure 3.5 provides a two-dimensional illustration of the proposed pose compensation concept to facilitate the explanation of the third step of the compensation procedure. In the left-hand diagram, the manipulator and wire are shown before compensation, where point *A* represents the J6 flange, point *B* represents the intermediate wire point J7, and point *C* represents the origin of the actual TCP at J8. The wire geometry is represented by the

vectors $\overrightarrow{AB}$ and $\overrightarrow{BC}$. Since the terminal wire segment $\overrightarrow{BC}$ is not parallel to the normal vector of the socket surface, directly advancing the manipulator toward the socket while maintaining this uncompensated posture would cause the wire tip to collide with the socket surface and result in insertion failure.

Two different angles can be identified in the uncompensated configuration. The angle γ represents the local bending angle between the wire segments $\overrightarrow{AB}$ and $\overrightarrow{BC}$, whereas X represents the pose compensation angle defined with respect to the J6 flange and the overall geometric relationship between J6 and the actual TCP. Although the virtual TCP rotation is defined at the origin of the actual TCP in the mathematical model, the physical correction is ultimately realized through the motion of the manipulator joints, with the J6 flange driving the entire end structure. Therefore, directly using the local bending angle γ as the compensation angle does not account for the complete geometric relationship among points A , B , and C , and may consequently result in either overcompensation or undercompensation. For this reason, the pose compensation angle X is adopted instead of directly using the local bending angle γ .

As shown in the right-hand diagram, the imaginary TCP frame is defined by applying a rotation of $-X^\circ$ in the external model, after which inverse kinematics is used to calculate the corresponding six joint angles. The resulting compensated posture is represented by points A , B'' , and C'' . After compensation, the terminal wire segment is oriented parallel to the normal vector of the socket surface, thereby providing the required insertion orientation. However, completing the rotational compensation does not imply that the wire tip is already positioned directly in front of the

socket. The orientation compensation and the subsequent positioning motion are treated as two separate steps. Once the compensated posture has been established, the manipulator maintains this orientation and translates with respect to the world coordinate system until the wire tip reaches the designated pre-insertion position in front of the socket. Since this stage changes only the Cartesian position while maintaining the compensated orientation, the alignment of the terminal wire segment is preserved during the motion. The manipulator can then advance along the specified insertion direction to complete the insertion task. It should also be emphasized that the axes of the imaginary TCP frame are not required to coincide with the socket normal; instead, the purpose of the imaginary TCP is to generate a manipulator posture that aligns the physical terminal wire segment with the socket normal. In addition, the imaginary TCP frame exists only within the compensation model and does not imply that its origin is physically located at the socket immediately after the rotational compensation.

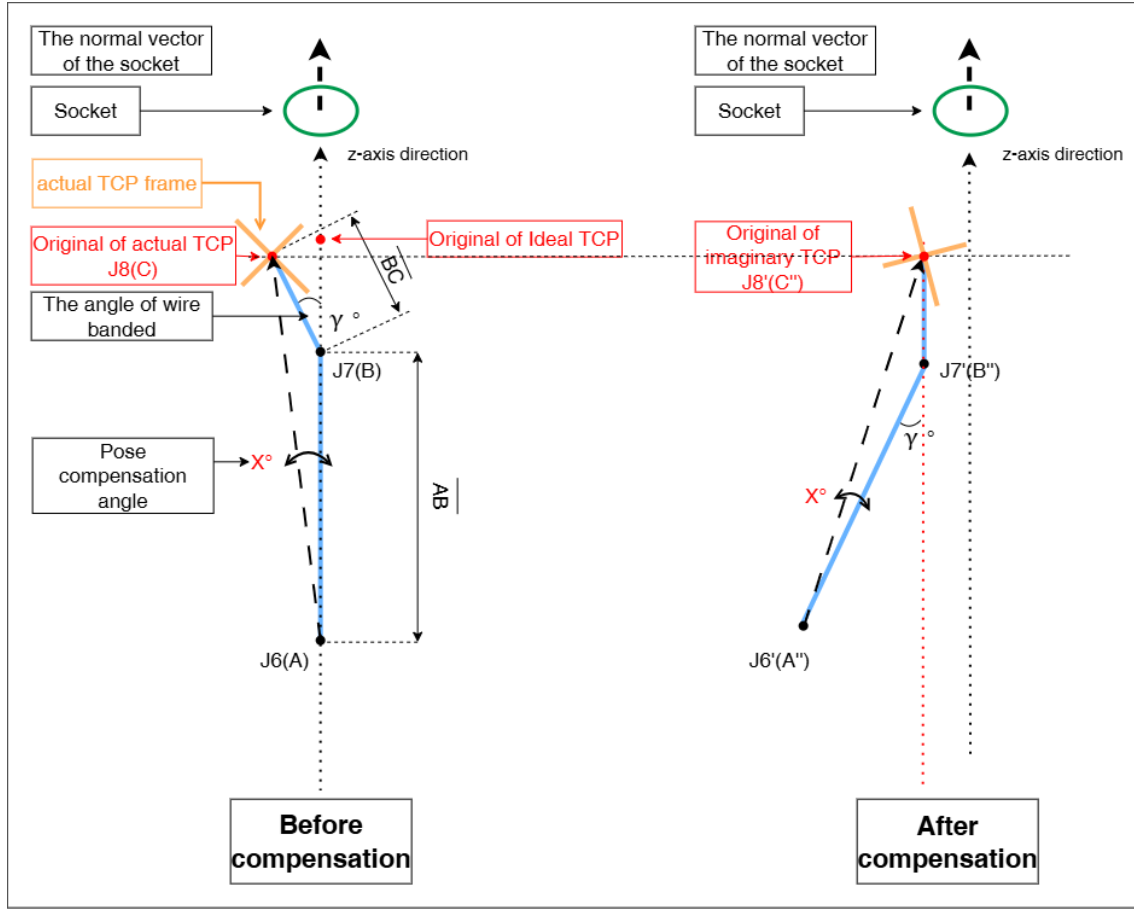
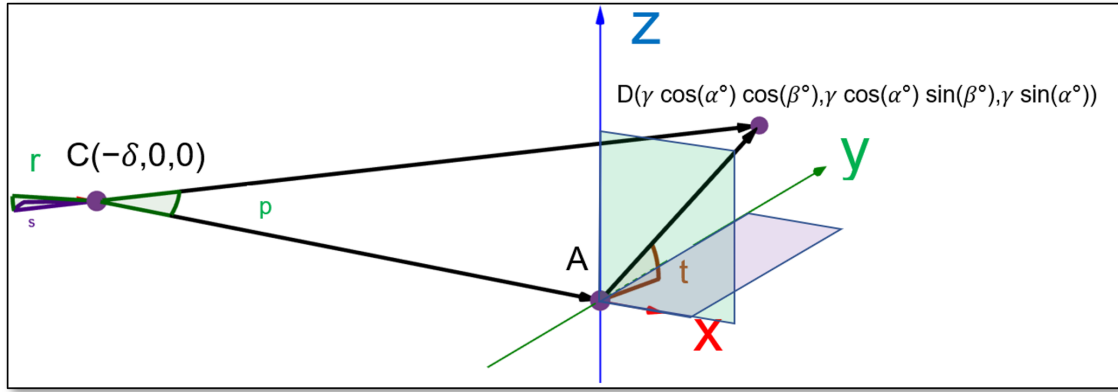
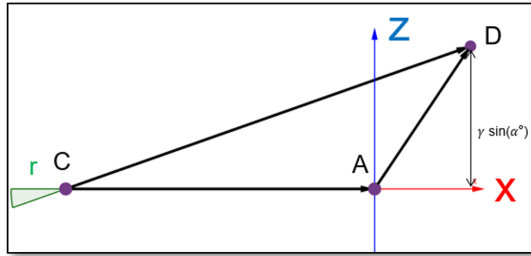


Figure 3.5: Concept for pose compensation in this research

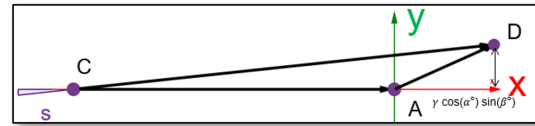
Next, regarding the implementation, this study utilizes the GeoGebra 3D calculator to visualize the vectors for explanation: As shown in the figure 3.6, under the 3D environment, three points exist in space after the manipulator grips the wire: point A, point B, and point C. The positions of point B and point C can vary depending on the gripper design and the wire product specification. Based on these three points in space, $\overline{AB}$, $\overline{BC}$, $\overline{AC}$, $\overrightarrow{AB}$, $\overrightarrow{BC}$, and $\overrightarrow{AC}$ can be obtained. The goal is to rotate $\overrightarrow{AC}$ in space to the position of $\overrightarrow{B''C''}$ shown in Figure 3.7, where $\overrightarrow{B''C''}$ aligns with the normal vector formed by sweeping a circle with a radius equal to the distance from point C to the x -axis, matching the concept proposed at the beginning of this section [27].



(a) Perspective view



(b) Side view (xz plane)



(c) Top view (xy plane)

Figure 3.8: The $\angle p$ can be decomposed into $\angle r$ and $\angle s$

First, to compensate for the decomposed $\angle s$, $\overrightarrow{AB}$ and $\overrightarrow{BC}$ need to rotate in different directions, as shown in the figure 3.9. According to the right-hand rule, with the thumb pointing along the z axis out of the page and the four fingers indicating a positive rotation direction, $\overrightarrow{AB}$ is configured to rotate in the positive direction about the z axis by s° , becoming $\overrightarrow{AB'}$. Meanwhile, $\overrightarrow{BC}$ is rotated in the negative direction about the z axis by s° , becoming $\overrightarrow{BC'}$. Mathematically, the rotation formulas are expressed as:

$$\overrightarrow{AB'} = R_z(s) * \overrightarrow{AB} = \begin{bmatrix} \cos(s) & -\sin(s) & 0 \\ \sin(s) & \cos(s) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \delta \\ 0 \\ 0 \end{bmatrix} \quad (3.11)$$

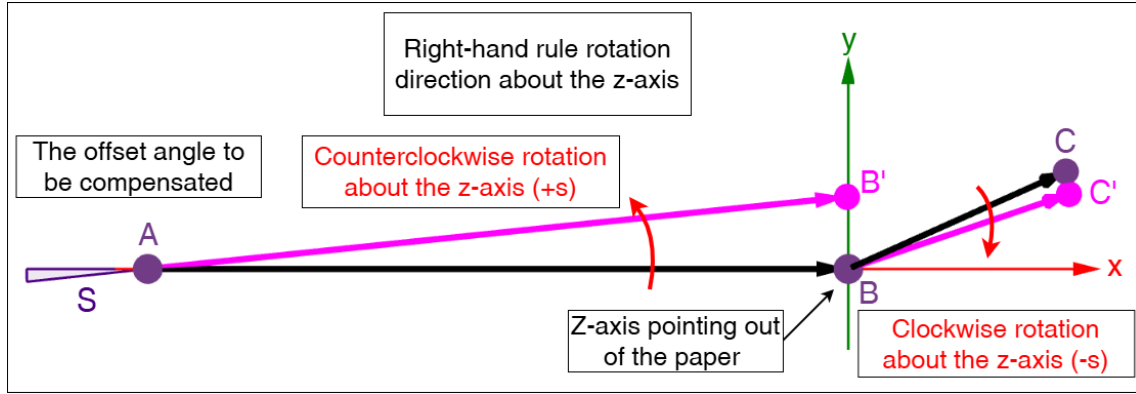


Figure 3.9: The $\angle s$ compensate way in this research

$$\overrightarrow{BC'} = R_z(-s) * \overrightarrow{BC} = \begin{bmatrix} \cos(-s) & -\sin(-s) & 0 \\ \sin(-s) & \cos(-s) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \gamma \cos(\alpha^\circ) \cos(\beta^\circ) \\ \gamma \cos(\alpha^\circ) \sin(\beta^\circ) \\ \gamma \sin(\alpha^\circ) \end{bmatrix} \quad (3.12)$$

R_z represents the rotation matrix about the z axis in 3D space, with $+s^\circ$ or $-s^\circ$ indicated inside the parentheses to correspond to the form of the rotation matrix. $\overrightarrow{AB}$ and $\overrightarrow{BC}$ are also expressed in matrix form, where the δ of $\overrightarrow{AB}$ is positive because $\overrightarrow{AB}$ is calculated as point B minus point A, cancelling out the negative sign.

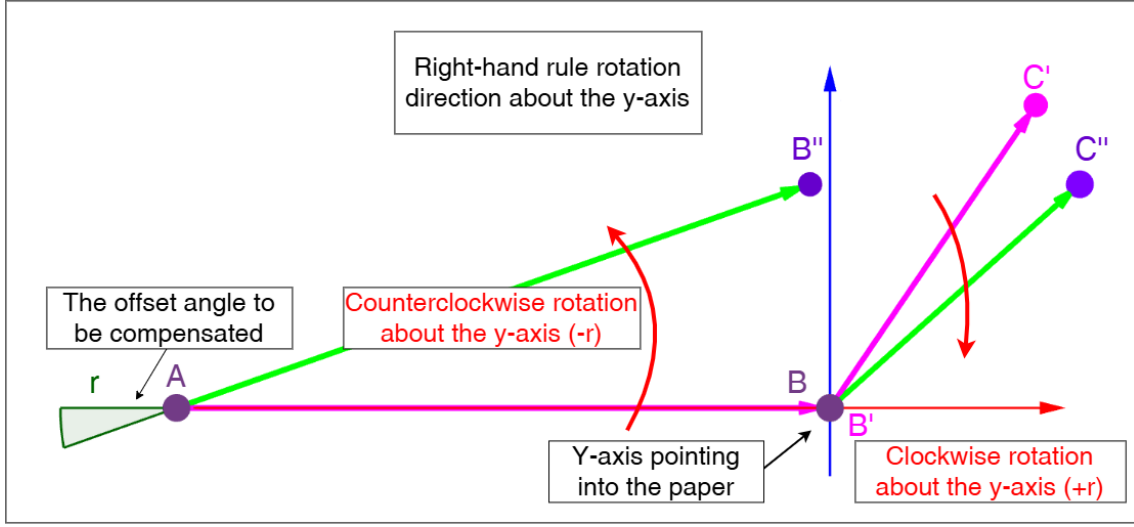


Figure 3.10: The $\angle r$ compensate way in this research

The concept for the second rotation is identical to the first. As shown in the figure 3.10, with the y axis pointing into the page, to compensate for $\angle r$ on the xz plane, $\overrightarrow{AB'}$ is rotated by $-r^\circ$, and $\overrightarrow{BC'}$ is rotated by $+r^\circ$. Expressing the two vector rotations in mathematical form yields:

$$\overrightarrow{AB''} = R_y(-r) * \overrightarrow{AB'} = \begin{bmatrix} \cos(-r) & 0 & \sin(-r) \\ 0 & 1 & 0 \\ -\sin(-r) & 0 & \cos(-r) \end{bmatrix} \begin{bmatrix} \overrightarrow{P_{BC'',x}} \\ \overrightarrow{P_{BC'',y}} \\ \overrightarrow{P_{BC'',z}} \end{bmatrix} = \overrightarrow{AB''} \quad (3.13)$$

After the two spatial rotations, as shown in the figure 3.11, to determine the endpoint coordinates of the two vectors $\overrightarrow{AB''}$ and $\overrightarrow{BC''}$ (points B'' and C''), point B'' is obtained by adding $\overrightarrow{AB''}$ to point A, and point C'' is obtained by adding $\overrightarrow{BC''}$ to point B:

$$A(-\delta, 0, 0) + \overrightarrow{AB''} = B''(B''_x, B''_y, B''_z) \quad (3.14)$$

$$B(0, 0, 0) + \overrightarrow{BC''} = C''(C''_x, C''_y, C''_z) \quad (3.15)$$

Finally, connecting point B'' and point C'' forms $\overrightarrow{B''C''}$, which aligns with

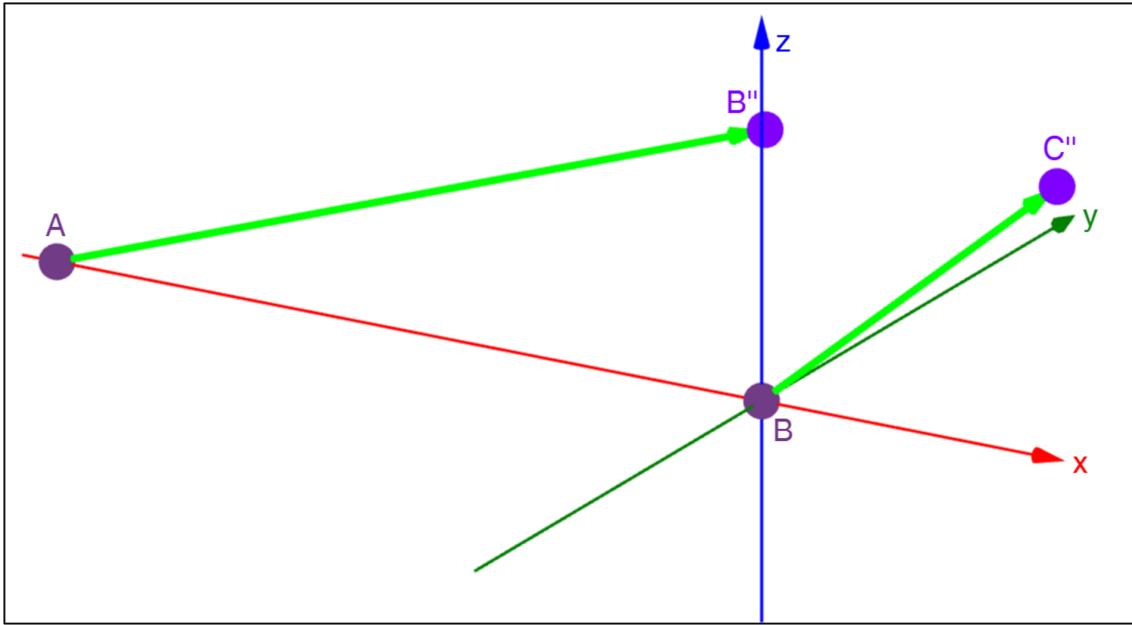


Figure 3.11: After the phase angle compensation, the phase would like this in the spatial

the normal vector of the circular surface formed by using the distance from point C to the x axis as the radius, as shown in the figure 3.12. After rotation, the wire tip vector represents $\overrightarrow{B''C''}$, and the normal vector of the circular surface represents the socket. The two align with each other, fulfilling the objective of this section. Note that the goal is only to align the wire tip vector with the normal vector, not to overlap them.

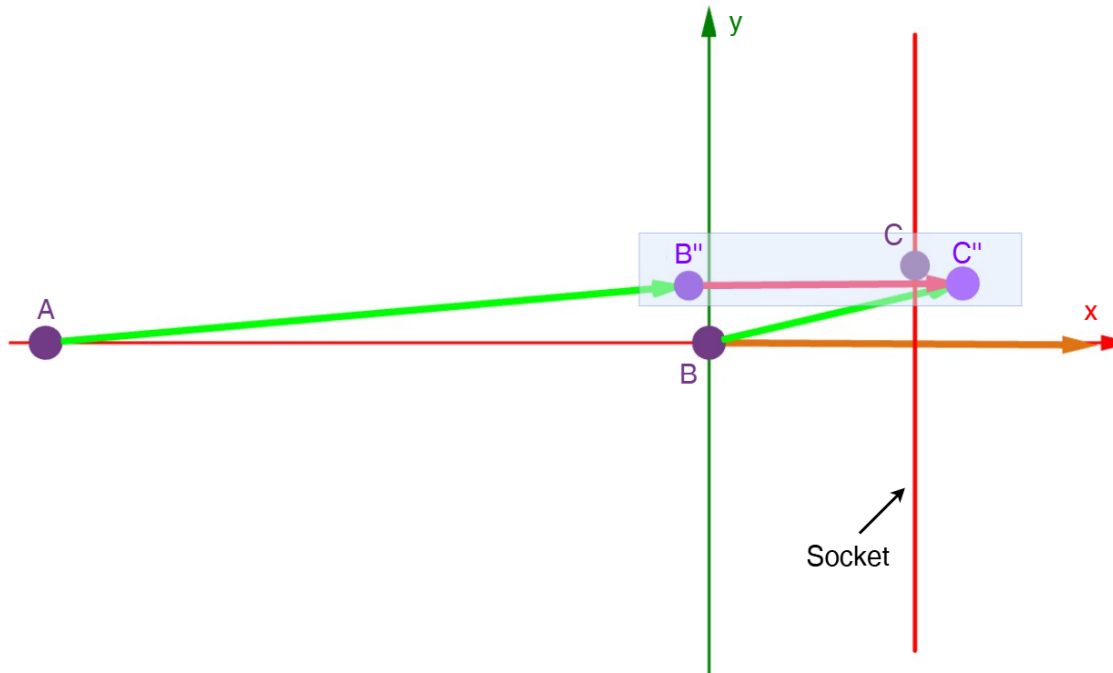


Figure 3.12: liking point G and D become a phase and it will align with circular's normal

3.4 Experiment Verification

This section explains how to apply the calculation method from the previous subsection to a physical environment. By aligning the wire with the socket and integrating MATLAB's inverse kinematics solver, the joint angles required to reach the position just before the insertion socket are computed based on the aligned wire and socket locations. Under this condition, the manipulator only needs to advance along a single axis to complete the insertion. This entire process can be executed without manual intervention, relying solely on MATLAB to derive the individual joint angles from the previously discussed mathematical model and inverse kinematics, which are then applied directly to the manipulator.

3.4.1 Calibration Procedure

To obtain the TCP transformation used in the physical experiment, a calibration-pointer-assisted five-point calibration procedure was performed using the ABB IRB 1090 manipulator controller. The tip of the wire held by the gripper was defined as the tool center point because it directly interacts with the target socket during insertion.

A fixed calibration pointer was installed within the manipulator workspace. The manipulator was then moved through several distinct orientations while maintaining contact between the wire tip and the same reference point of the calibration pointer. These recorded poses allowed the controller to estimate the translational offset of the wire-tip TCP relative to the manipulator flange.

Additional orientation points were used to define the direction of the tool coordinate frame. After the five-point procedure was completed, the controller generated the TCP position, coordinate-frame orientation, and calibration-error information. The same procedure was repeated for each wire-orientation condition, including the R, L, U, and D groups. The resulting TCP data were subsequently used as the tool transformations in the MATLAB simulation.

3.4.2 Gripper finger Design In Experiment

The gripper fingers in this study is to provide a controlled and repeatable wire-holding condition for verifying the proposed pose compensation model. As shown in Figure 3.13, the gripper fingers were intentionally de-

signed with a 15° leftward deflection to emulate the angular deviation that may occur when a wire is picked up by the manipulator. In addition, the gripper fingers include a sleeve structure that encloses and fixes the wire, increasing the contact friction and preventing the wire from moving during the insertion process. In this experiment, the center of the sleeve was designed to coincide with the rotational center of the manipulator's J6 axis. Therefore, if the wire is picked up with an incorrect facing direction, its orientation can be corrected simply by rotating J6, allowing the wire to return to the proper insertion direction.

In the physical experiments, due to the limited bending range of the wire and to simplify the experimental design, a camera was not integrated to capture and detect the actual wire bending angle. Instead, by fixing the relative angle between the gripper fingers and the gripper body, the bending condition generated during wire grasping can be approximated in a repeatable manner. Taking the 15° leftward deflection of the gripper fingers as the experimental example, the joint angles calculated in the virtual environment are used to compensate for this angular offset, ensuring that the wire tip faces the socket at a 0° orientation. It is worth noting that this 15° does not represent the angle between the manipulator's J6 axis and the wire tip, but rather the angle between the gripper tool point and the wire tip.

3.4.3 Wire fixture In Experiment

The wire fixture in this study is to provide a stable and repeatable initial position for the wire before each insertion routine. Since the wire must be protected from environmental disturbances during the experiment, the

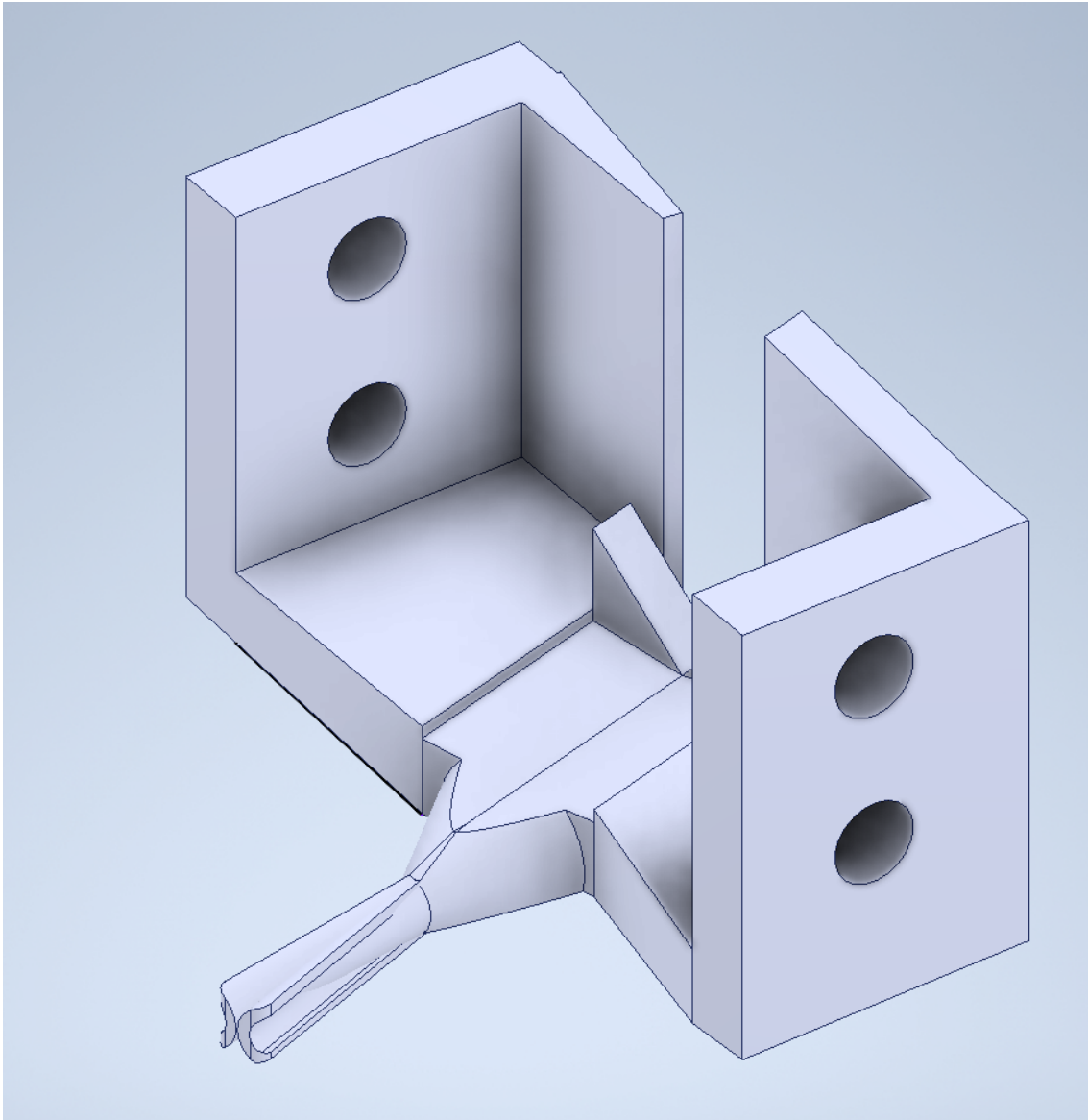


Figure 3.13: Gripper finger design in this research

fixture is used to constrain its position and orientation. In addition, the wire has distinct top and bottom sides; therefore, if the wire is oriented in the wrong direction, it cannot be inserted into the wire socket. To address this issue, a wire fixture was designed and fabricated using 3D printing, as shown in Figure 3.14.

After the wire is manually placed into the fixture, its initial pose can

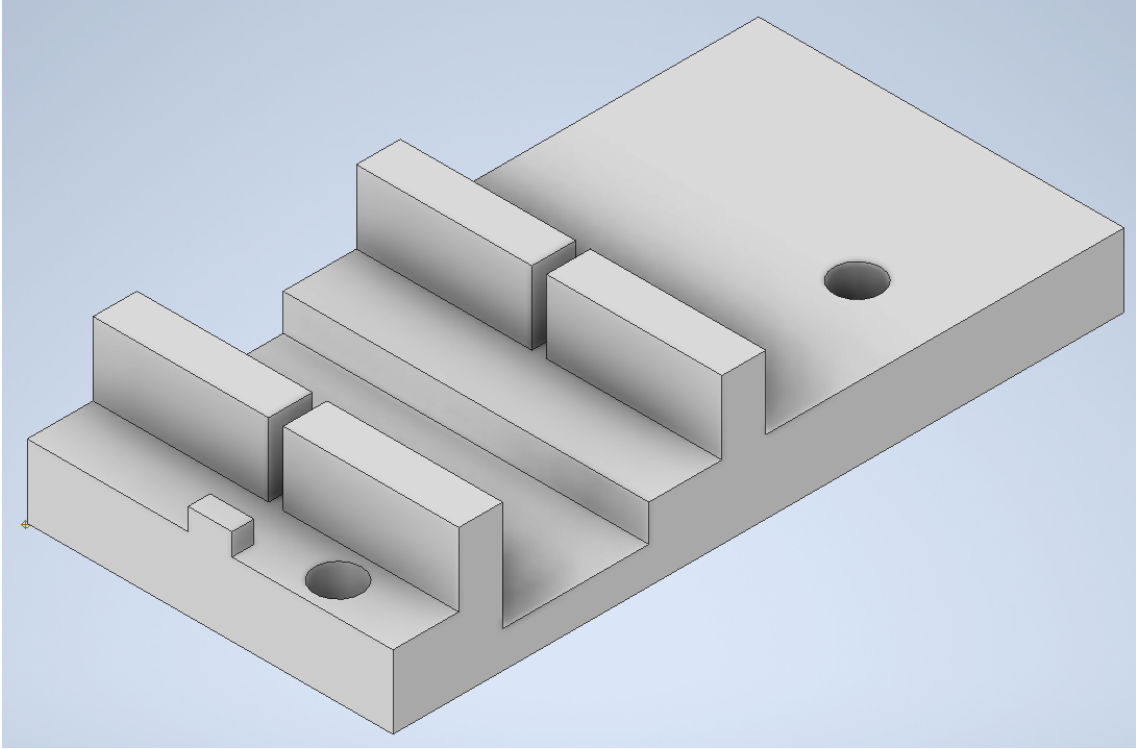


Figure 3.14: Wire fixture design in this research

be maintained consistently before being grasped by the manipulator. The intended gripping position for the gripper fingers is then pre-programmed into the manipulator controller. Consequently, during each insertion routine, the manipulator automatically picks up the wire from the wire fixture. This fixture design ensures that the wire is presented to the gripper under a controlled initial condition, allowing the subsequent insertion experiment to focus on the pose compensation performance of the proposed MATLAB model.

3.4.4 Wire Socket

The feature of the wire socket in this study is the aperture diameter. The wire socket refers to the target aperture configuration used in the insertion

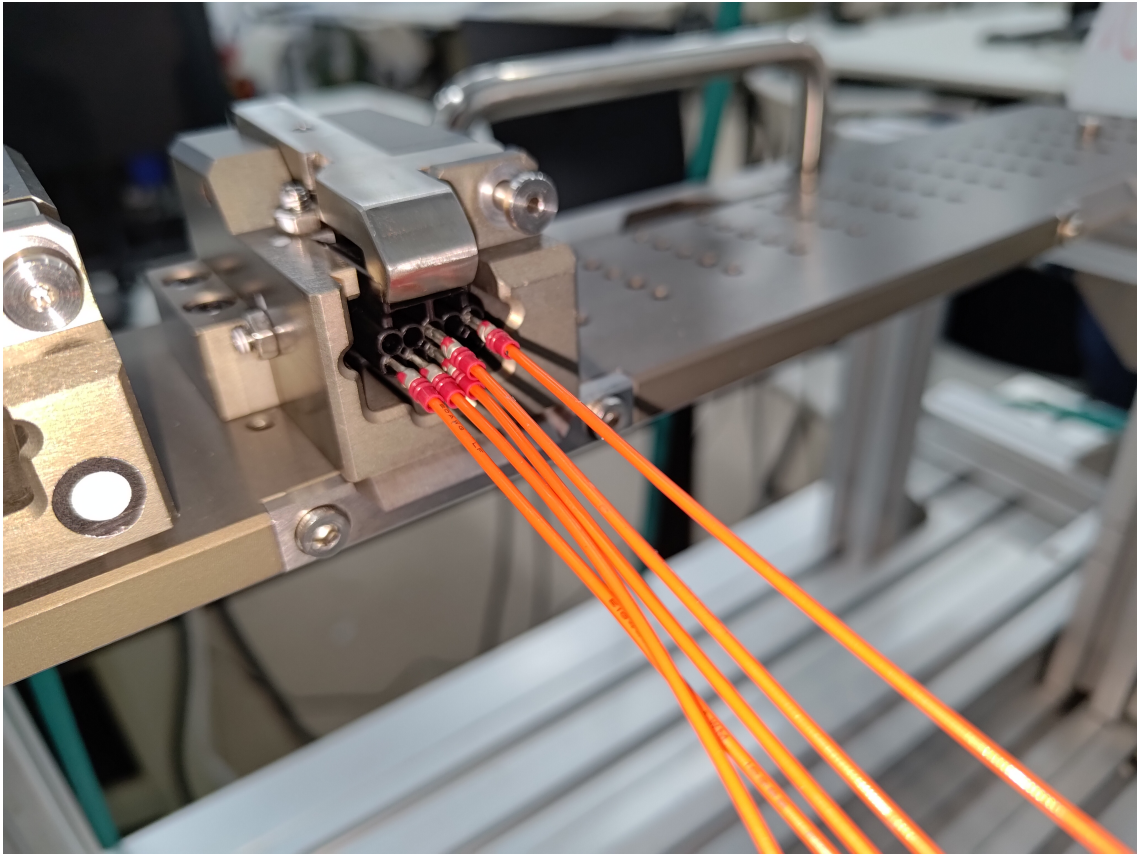


Figure 3.15: Wire socket in this research

experiment, which consists of ten sockets, as shown in Figure 3.15. Each socket has an aperture diameter of 3 mm, while the wire tip diameter is 0.7 mm and have 10 socket in one connector.

It should be noted that the levelness of the socket platform also affects the coordinate position of each socket during insertion. If the platform is not perfectly horizontal, slight positional deviations may occur among the sockets, and every other socket may introduce an additional small offset. Therefore, both the aperture diameter and the platform condition must be considered when evaluating the insertion accuracy.

Chapter 4 Results And Discussion

This chapter discusses the achievements realized in this study along with some mathematical details of the experiments, proving the accuracy and validity of the experimental results from a mathematical perspective.

4.1 Data Preparation

Tables 4.1 and 4.2 summarize the key spatial data required for the MATLAB simulation and subsequent inverse kinematics computation. In addition to the hardware configuration, the simulation requires two main types of information: the calibrated TCP data of each tool condition and the corresponding target pose of the socket. Therefore, before constructing the simulation model, this study first records the TCP information and the target rotarget data used in the physical insertion experiment.

The TCP data presented in Table 4.1 were obtained using the calibration-pointer-assisted five-point calibration procedure described in Section 3.4.1. In this table, the tool names, such as R, L, U, and D, represent different tool or wire-orientation conditions used in the experiment. The columns Max Error, Min Error, and Mean Error describe the calibration error range of each TCP. The parameters X, Y, and Z represent the translational offset of the TCP, while Rx, Ry, and Rz represent the rotational orientation of the TCP coordinate frame.

Table 4.2 lists the corresponding target poses of the socket recorded under the different TCP conditions. Since different TCP definitions affect

how the manipulator controller records and interprets the target pose during teaching, each target point is recorded together with its corresponding TCP condition to maintain consistency between the physical experiment and the MATLAB simulation. For example, the target point RTR_1 corresponds to the TCP condition R_1, and the same naming rule is applied to the other experimental groups.

In Table 4.2, X, Y, and Z represent the Cartesian position of the socket in millimeters. The parameters q1 to q4 are the quaternion components used to describe the orientation of the target frame, rather than the joint angles of the manipulator. The parameters cf1, cf4, cf6, and cfx are the ABB configuration data stored in the robtarget, which describe the manipulator configuration used when the target was recorded. These configuration values help ensure that the inverse kinematics solution corresponds to the same physical posture used in the experiment. Together, Tables 4.1 and 4.2 provide the necessary relationship between each TCP condition and its corresponding socket target pose, forming the data foundation for the MATLAB-based pose compensation and inverse kinematics calculation.

4.2 Simulation in Matlab

This section explains the MATLAB simulation workflow used to compute the compensated joint angles for wire insertion. The model requires three main inputs: the target pose of the socket, the original TCP obtained from calibration, and the angular deviation caused by the wire banded. Based on these inputs, the MATLAB model constructs a compensated imaginary TCP to represent the required wire-tip orientation before insertion.

Table 4.1: Tool Errors and Spatial Transformations Analysis

Tool name	Max (mm)	Min (mm)	Mean (mm)	X (mm)	Y (mm)	Z (mm)	Rx (deg)	Ry (deg)	Rz (deg)
R_1	0.399	0.331	0.382	-0.149	-4.984	81.592	-0.024	50.047	0
R_2	0.342	0.288	0.318	-6.050	-6.720	72.688	-179.992	88.285	-180
R_3	1.220	0.553	0.956	2.565	-6.067	76.390	179.999	88.691	180
L_1	0.797	0.604	0.744	1.136	5.951	81.994	-179.995	89.053	-180
L_2	0.781	0.295	0.574	1.774	5.841	83.649	-0.004	88.937	0
L_3	0.565	0.196	0.422	1.520	5.119	81.819	-179.998	86.742	180
U_1	0.484	0.144	0.379	-1.982	-0.496	77.527	0.006	84.358	0
U_2	0.784	0.673	0.741	-2.195	-1.844	77.855	-0.002	87.227	0
U_3	0.573	0.477	0.537	-8.777	-0.870	69.182	0.022	84.346	0
D_1	0.957	0.313	0.796	8.959	-1.391	75.646	-179.869	80.467	180
D_2	0.518	0.298	0.459	9.007	-1.925	76.855	-179.998	81.349	-180
D_3	0.685	0.151	0.438	14.165	-2.930	80.023	-179.993	75.415	180

Table 4.2: Target Spatial Positions and Orientations Data

target name	X	Y	Z	q1	q2	q3	q4	cf1	cf4	cf6	cdx
RTR_1	511.40	48.31	216.92	0.363149	-0.124863	0.922172	0.046162	0	-1	0	1
RTR_2	502.18	44.59	222.14	0.002548	0.130629	-0.991428	-0.000292	0	-1	0	1
RTR_3	506.82	45.92	213.94	0.014952	-0.132984	0.991005	-0.000933	0	-1	0	1
RTL_1	512.88	47.10	217.08	0.000461	4.58×10^{-5}	-1.000000	2.92×10^{-5}	0	0	-1	1
RTL_2	514.55	46.99	216.47	0.017077	-4.84×10^{-5}	0.999854	5.37×10^{-5}	0	0	-1	1
RTL_3	512.71	46.26	216.69	0.020625	4.77×10^{-5}	-0.999787	7.46×10^{-6}	0	0	-1	1
RTU_1	508.32	46.61	214.91	0.029213	4.19×10^{-5}	-0.999573	-1.66×10^{-5}	0	0	-1	1
RTU_2	508.68	45.26	215.07	0.054229	4.20×10^{-5}	-0.998528	-0.000093	0	0	-1	1
RTU_3	501.14	46.23	222.93	0.029111	4.59×10^{-5}	-0.999576	0.000123	0	0	-1	1
RTD_1	509.44	45.64	216.02	0.060721	-0.028917	0.997715	-0.006504	0	0	-1	1
RTD_2	510.64	45.18	216.33	0.068374	-0.029022	0.997224	-0.005164	0	0	-1	1
RTD_3	515.19	44.38	212.29	0.016657	-0.028719	0.999426	-0.006703	0	0	-1	1

It should be emphasized that the MATLAB implementation is slightly different from the vector-rotation explanation presented in Chapter 3. In Chapter 3, the compensation concept is illustrated by rotating vectors to align the wire direction with the normal vector of the socket. However, in the physical experiment, the wire is fixed by the gripper and cannot be

independently rotated as a separate object. Therefore, in the MATLAB model, the same compensation effect is achieved by rotating the original TCP coordinate frame about its own axis. Although the implementation method differs from the geometric vector explanation, the applied rotation angle and the resulting compensated orientation are equivalent. Therefore, the final compensation result is not affected.

After the angular deviation is import into the model, MATLAB performs inverse kinematics to solve the corresponding joint angles of the six-axis manipulator. In other words, the model does not directly command the manipulator to rotate the gripper by the compensation angle. Instead of directly commanding the physical manipulator to rotate the gripper by the compensation angle, the MATLAB model first modifies the TCP definition by adding the wire-deflection angle. The inverse kinematics solver then calculates the six joint angles required for this modified TCP to align with the target socket frame. Therefore, the resulting joint angles already contain the pose correction needed to compensate for the wire deflection.

It should also be noted that the rotation applied to the TCP model must be interpreted inversely when observing the physical motion of the manipulator. For example, although the required physical correction of the gripper may correspond to a positive rotation about the TCP Z-axis, the MATLAB model applies a negative rotation to the tool transformation. This is because the rotation is used to redefine the TCP frame not to directly command the physical motion. When the modified TCP frame is constrained to align with the target frame through inverse kinematics, the original TCP or gripper frame exhibits an equivalent opposite compensation effect.

Therefore, the complete simulation process can be summarized as follows: the target pose, ideal TCP, and wire angular deviation are first entered into the MATLAB model; the model then generates a compensated imaginary TCP frame by rotating the original TCP coordinate frame; inverse kinematics is applied to solve the required manipulator posture; finally, the six joint angles are obtained. These calculated joint angles represent the physical manipulator configuration needed to place the wire tip in front of the socket while simultaneously correcting the angular deviation of the wire.

Table 4.3 summarizes the simulation results obtained by combining different target points with different TCP configurations. This table records the experimental and simulation data for each wire insertion trial. The columns from ID to Tool_Name describe the socket index, and TCP setting. The columns X_mm, Y_mm, and Z_mm represent the Cartesian position of the target or TCP in millimeters. The columns J1_deg to J6_deg record the corresponding joint angles of the six-axis manipulator. Finally, PosErr_mm is used to evaluate the position errors between the calculated TCP pose and the desired target pose.

Table 4.3: Complete Dataset of Inverse Kinematics Verification (120 Points)

ID	Tool Name	X (mm)	Y (mm)	Z (mm)	J1 (deg)	J2 (deg)	J3 (deg)	J4 (deg)	J5 (deg)	J6 (deg)	PosErr (mm)
1	R_1	509.47	45.86	213.05	6.27	55.06	12.54	-0.09	-69.12	-1.18	4.2×10^{-8}
2	R_1	509.47	42.86	213.05	5.78	55.03	12.61	-0.60	-69.17	-0.98	4.1×10^{-8}
3	R_1	509.47	39.86	213.05	5.30	55.01	12.69	-1.13	-69.22	-0.79	4.0×10^{-8}
4	R_1	509.47	36.86	213.05	4.81	54.98	12.75	-1.65	-69.27	-0.59	4.0×10^{-8}
Next page											

Table 4.3: Complete Dataset of Inverse Kinematics Verification (120 Points)

ID	Tool Name	X (mm)	Y (mm)	Z (mm)	J1 (deg)	J2 (deg)	J3 (deg)	J4 (deg)	J5 (deg)	J6 (deg)	PosErr (mm)
5	R_1	509.47	33.86	213.05	4.33	54.96	12.81	-2.17	-69.31	-0.40	4.0×10^{-8}
6	R_1	509.47	45.86	210.05	6.27	55.63	12.30	-0.09	-69.45	-1.18	4.0×10^{-8}
7	R_1	509.47	42.86	210.05	5.78	55.60	12.38	-0.61	-69.51	-0.99	3.9×10^{-8}
8	R_1	509.47	39.86	210.05	5.29	55.58	12.45	-1.12	-69.55	-0.79	3.9×10^{-8}
9	R_1	509.47	36.86	210.05	4.81	55.55	12.51	-1.64	-69.60	-0.60	3.9×10^{-8}
10	R_1	509.47	33.86	210.05	4.33	55.53	12.57	-2.16	-69.65	-0.41	3.9×10^{-8}
1	R_2	529.56	50.92	190.69	6.90	55.04	12.59	0.60	-69.14	-1.25	5.1×10^{-8}
2	R_2	529.56	47.92	190.69	6.41	55.01	12.67	0.08	-69.19	-1.05	4.9×10^{-8}
3	R_2	529.56	44.92	190.69	5.93	54.98	12.75	-0.44	-69.24	-0.85	4.8×10^{-8}
4	R_2	529.56	41.92	190.69	5.44	54.96	12.83	-0.95	-69.29	-0.66	4.8×10^{-8}
5	R_2	529.56	38.92	190.69	4.95	54.93	12.89	-1.47	-69.34	-0.46	4.7×10^{-8}
6	R_2	529.56	50.92	187.69	6.90	55.61	12.35	0.60	-69.47	-1.24	4.9×10^{-8}
7	R_2	529.56	47.92	187.69	6.41	55.58	12.44	0.08	-69.52	-1.05	4.8×10^{-8}
8	R_2	529.56	44.92	187.69	5.93	55.55	12.51	-0.43	-69.58	-0.86	4.7×10^{-8}
9	R_2	529.56	41.92	187.69	5.44	55.53	12.59	-0.95	-69.63	-0.66	4.6×10^{-8}
10	R_2	529.56	38.92	187.69	4.95	55.50	12.65	-1.47	-69.67	-0.47	4.6×10^{-8}
1	R_3	511.25	46.34	215.66	6.33	55.05	12.53	-0.16	-69.11	-1.16	4.2×10^{-8}
2	R_3	511.25	43.34	215.66	5.84	55.03	12.61	-0.68	-69.16	-0.96	4.1×10^{-8}
3	R_3	511.25	40.34	215.66	5.35	55.00	12.68	-1.20	-69.21	-0.77	4.0×10^{-8}
4	R_3	511.25	37.34	215.66	4.87	54.98	12.75	-1.71	-69.26	-0.57	4.0×10^{-8}
5	R_3	511.25	34.34	215.66	4.38	54.95	12.81	-2.23	-69.30	-0.38	4.0×10^{-8}
6	R_3	511.25	46.34	212.66	6.33	55.62	12.29	-0.16	-69.44	-1.16	4.1×10^{-8}
7	R_3	511.25	43.34	212.66	5.84	55.60	12.37	-0.68	-69.49	-0.97	4.0×10^{-8}
8	R_3	511.25	40.34	212.66	5.35	55.57	12.44	-1.19	-69.54	-0.77	4.0×10^{-8}
9	R_3	511.25	37.34	212.66	4.87	55.55	12.50	-1.71	-69.59	-0.58	3.9×10^{-8}
10	R_3	511.25	34.34	212.66	4.38	55.52	12.56	-2.23	-69.63	-0.39	3.9×10^{-8}
1	L_1	512.88	46.60	216.08	2.59	54.89	12.13	-7.95	-68.16	3.49	9.8×10^{-8}
2	L_1	512.88	43.60	216.08	2.59	54.89	12.13	-7.95	-68.16	3.49	9.4×10^{-8}
3	L_1	512.88	40.60	216.08	2.59	54.89	12.13	-7.95	-68.16	3.49	8.1×10^{-8}
4	L_1	512.88	37.60	216.08	2.59	54.89	12.13	-7.95	-68.16	3.49	7.2×10^{-8}
5	L_1	512.88	34.60	216.08	2.59	54.89	12.13	-7.95	-68.16	3.49	7.1×10^{-8}
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Table 4.3: Complete Dataset of Inverse Kinematics Verification (120 Points)

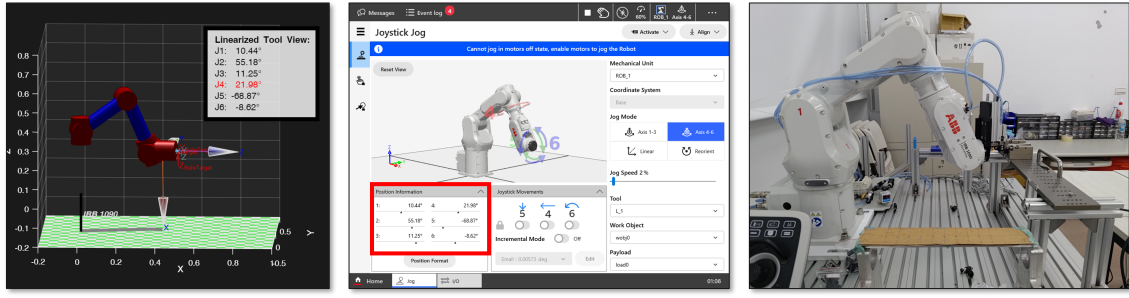
ID	Tool Name	X (mm)	Y (mm)	Z (mm)	J1 (deg)	J2 (deg)	J3 (deg)	J4 (deg)	J5 (deg)	J6 (deg)	PosErr (mm)
6	L_1	512.88	46.60	213.08	2.59	54.89	12.13	-7.95	-68.16	3.49	9.3×10^{-8}
7	L_1	512.88	43.60	213.08	2.59	54.89	12.13	-7.95	-68.16	3.49	8.9×10^{-8}
8	L_1	512.88	40.60	213.08	2.59	54.89	12.13	-7.95	-68.16	3.49	7.6×10^{-8}
9	L_1	512.88	37.60	213.08	2.59	54.89	12.13	-7.95	-68.16	3.49	7.3×10^{-8}
10	L_1	512.88	34.60	213.08	2.59	54.89	12.13	-7.95	-68.16	3.49	6.8×10^{-8}
1	L_2	514.55	46.99	216.47	2.56	54.88	12.14	-8.01	-68.15	3.53	9.5×10^{-8}
2	L_2	514.55	43.99	216.47	2.56	54.88	12.14	-8.01	-68.15	3.53	8.8×10^{-8}
3	L_2	514.55	40.99	216.47	2.56	54.88	12.14	-8.01	-68.15	3.53	7.3×10^{-8}
4	L_2	514.55	37.99	216.47	2.56	54.88	12.14	-8.01	-68.15	3.53	5.9×10^{-8}
5	L_2	514.55	34.99	216.47	2.56	54.88	12.14	-8.01	-68.15	3.53	5.5×10^{-8}
6	L_2	514.55	46.99	213.47	2.56	54.88	12.14	-8.01	-68.15	3.53	9.0×10^{-8}
7	L_2	514.55	43.99	213.47	2.56	54.88	12.14	-8.01	-68.15	3.53	8.3×10^{-8}
8	L_2	514.55	40.99	213.47	2.56	54.88	12.14	-8.01	-68.15	3.53	6.8×10^{-8}
9	L_2	514.55	37.99	213.47	2.56	54.88	12.14	-8.01	-68.15	3.53	5.7×10^{-8}
10	L_2	514.55	34.99	213.47	2.56	54.88	12.14	-8.01	-68.15	3.53	5.1×10^{-8}
1	L_3	512.71	46.26	216.69	2.57	54.88	12.14	-8.00	-68.15	3.53	8.2×10^{-8}
2	L_3	512.71	43.26	216.69	2.57	54.88	12.14	-8.00	-68.15	3.53	9.4×10^{-8}
3	L_3	512.71	40.26	216.69	2.57	54.88	12.14	-8.00	-68.15	3.53	9.7×10^{-8}
4	L_3	512.71	37.26	216.69	2.57	54.88	12.14	-8.00	-68.15	3.53	9.4×10^{-8}
5	L_3	512.71	34.26	216.69	2.57	54.88	12.14	-8.00	-68.15	3.53	8.7×10^{-8}
6	L_3	512.71	46.26	213.69	2.57	54.88	12.14	-8.00	-68.15	3.53	8.6×10^{-8}
7	L_3	512.71	43.26	213.69	2.57	54.88	12.14	-8.00	-68.15	3.53	9.1×10^{-8}
8	L_3	512.71	40.26	213.69	2.57	54.88	12.14	-8.00	-68.15	3.53	9.2×10^{-8}
9	L_3	512.71	37.26	213.69	2.57	54.88	12.14	-8.00	-68.15	3.53	8.8×10^{-8}
10	L_3	512.71	34.26	213.69	2.57	54.88	12.14	-8.00	-68.15	3.53	8.3×10^{-8}
1	U_1	508.32	46.61	214.91	6.20	54.87	12.17	-0.19	-68.13	-1.09	2.0×10^{-9}
2	U_1	508.32	43.61	214.91	5.72	54.85	12.24	-0.71	-68.19	-0.89	1.9×10^{-9}
3	U_1	508.32	40.61	214.91	5.23	54.83	12.31	-1.23	-68.24	-0.69	1.7×10^{-9}
4	U_1	508.32	37.61	214.91	4.74	54.81	12.37	-1.74	-68.29	-0.50	1.5×10^{-9}
5	U_1	508.32	34.61	214.91	4.25	54.79	12.42	-2.26	-68.33	-0.30	1.3×10^{-9}
6	U_1	508.32	46.61	211.91	6.20	55.45	11.92	-0.19	-68.47	-1.09	2.0×10^{-9}
Next page											

Table 4.3: Complete Dataset of Inverse Kinematics Verification (120 Points)

ID	Tool Name	X (mm)	Y (mm)	Z (mm)	J1 (deg)	J2 (deg)	J3 (deg)	J4 (deg)	J5 (deg)	J6 (deg)	PosErr (mm)
7	U_1	508.32	43.61	211.91	5.72	55.42	11.99	-0.71	-68.52	-0.89	1.9×10^{-9}
8	U_1	508.32	40.61	211.91	5.23	55.40	12.06	-1.23	-68.57	-0.70	1.7×10^{-9}
9	U_1	508.32	37.61	211.91	4.74	55.37	12.12	-1.74	-68.61	-0.50	1.4×10^{-9}
10	U_1	508.32	34.61	211.91	4.25	55.35	12.18	-2.26	-68.65	-0.31	1.2×10^{-9}
1	U_2	508.68	45.26	215.07	6.07	54.88	12.14	-0.21	-68.14	-1.07	2.0×10^{-9}
2	U_2	508.68	42.26	215.07	5.58	54.86	12.21	-0.72	-68.19	-0.88	1.9×10^{-9}
3	U_2	508.68	39.26	215.07	5.09	54.83	12.28	-1.24	-68.24	-0.68	1.7×10^{-9}
4	U_2	508.68	36.26	215.07	4.60	54.81	12.34	-1.75	-68.29	-0.48	1.5×10^{-9}
5	U_2	508.68	33.26	215.07	4.11	54.79	12.40	-2.26	-68.33	-0.29	1.3×10^{-9}
6	U_2	508.68	45.26	212.07	6.07	55.45	11.90	-0.21	-68.47	-1.07	2.0×10^{-9}
7	U_2	508.68	42.26	212.07	5.58	55.42	11.97	-0.73	-68.52	-0.88	1.9×10^{-9}
8	U_2	508.68	39.26	212.07	5.09	55.40	12.03	-1.24	-68.57	-0.68	1.7×10^{-9}
9	U_2	508.68	36.26	212.07	4.60	55.37	12.09	-1.75	-68.61	-0.49	1.5×10^{-9}
10	U_2	508.68	33.26	212.07	4.11	55.35	12.15	-2.26	-68.65	-0.29	1.2×10^{-9}
1	U_3	501.14	46.23	222.93	6.26	55.05	12.53	-0.09	-69.12	-1.18	1.7×10^{-9}
2	U_3	501.14	43.23	222.93	5.78	55.03	12.61	-0.61	-69.17	-0.98	1.6×10^{-9}
3	U_3	501.14	40.23	222.93	5.29	55.00	12.68	-1.13	-69.22	-0.79	1.4×10^{-9}
4	U_3	501.14	37.23	222.93	4.81	54.98	12.75	-1.65	-69.26	-0.59	1.3×10^{-9}
5	U_3	501.14	34.23	222.93	4.32	54.95	12.81	-2.17	-69.30	-0.39	1.1×10^{-9}
6	U_3	501.14	46.23	219.93	6.26	55.62	12.29	-0.09	-69.45	-1.18	1.7×10^{-9}
7	U_3	501.14	43.23	219.93	5.78	55.60	12.37	-0.61	-69.50	-0.99	1.6×10^{-9}
8	U_3	501.14	40.23	219.93	5.29	55.57	12.44	-1.12	-69.55	-0.79	1.4×10^{-9}
9	U_3	501.14	37.23	219.93	4.81	55.55	12.50	-1.71	-69.59	-0.59	1.2×10^{-9}
10	U_3	501.14	34.23	219.93	4.32	55.52	12.56	-2.23	-69.63	-0.40	1.1×10^{-9}
1	D_1	509.44	45.64	216.02	6.07	66.15	3.92	2.15	-96.56	-1.73	1.2×10^{-9}
2	D_1	509.44	42.64	216.02	5.58	66.11	3.99	1.62	-96.61	-1.54	1.2×10^{-9}
3	D_1	509.44	39.64	216.02	5.09	66.07	4.05	1.09	-96.65	-1.35	1.2×10^{-9}
4	D_1	509.44	36.64	216.02	4.60	66.04	4.12	0.55	-96.70	-1.15	1.2×10^{-9}
5	D_1	509.44	33.64	216.02	4.11	66.00	4.18	0.01	-96.75	-0.96	1.2×10^{-9}
6	D_1	509.44	45.64	213.02	6.07	66.86	3.39	2.14	-96.96	-1.74	1.2×10^{-9}
7	D_1	509.44	42.64	213.02	5.58	66.82	3.46	1.61	-97.02	-1.54	1.2×10^{-9}
Next page											

Table 4.3: Complete Dataset of Inverse Kinematics Verification (120 Points)

ID	Tool Name	X (mm)	Y (mm)	Z (mm)	J1 (deg)	J2 (deg)	J3 (deg)	J4 (deg)	J5 (deg)	J6 (deg)	PosErr (mm)
8	D_1	509.44	39.64	213.02	5.09	66.78	3.52	1.08	-97.06	-1.35	1.2×10^{-9}
9	D_1	509.44	36.64	213.02	4.60	66.75	3.59	0.54	-97.11	-1.16	1.2×10^{-9}
10	D_1	509.44	33.64	213.02	4.11	66.71	3.65	0.00	-97.16	-0.96	1.2×10^{-9}
1	D_2	510.64	45.18	216.33	6.07	66.15	3.92	2.15	-96.56	-1.73	1.2×10^{-9}
2	D_2	510.64	42.18	216.33	5.58	66.11	3.99	1.62	-96.61	-1.54	1.2×10^{-9}
3	D_2	510.64	39.18	216.33	5.09	66.07	4.05	1.09	-96.65	-1.35	1.2×10^{-9}
4	D_2	510.64	36.18	216.33	4.60	66.04	4.12	0.55	-96.70	-1.15	1.2×10^{-9}
5	D_2	510.64	33.18	216.33	4.11	66.00	4.18	0.01	-96.75	-0.96	1.3×10^{-9}
6	D_2	510.64	45.18	213.33	6.07	66.86	3.39	2.14	-96.96	-1.74	1.2×10^{-9}
7	D_2	510.64	42.18	213.33	5.58	66.82	3.46	1.61	-97.02	-1.54	1.2×10^{-9}
8	D_2	510.64	39.18	213.33	5.09	66.78	3.52	1.08	-97.06	-1.35	1.2×10^{-9}
9	D_2	510.64	36.18	213.33	4.60	66.75	3.59	0.54	-97.11	-1.16	1.2×10^{-9}
10	D_2	510.64	33.18	213.33	4.11	66.71	3.65	0.00	-97.16	-0.96	1.3×10^{-9}
1	D_3	515.19	44.38	212.29	6.07	66.15	3.92	2.15	-96.56	-1.73	1.2×10^{-9}
2	D_3	515.19	41.38	212.29	5.58	66.11	3.99	1.62	-96.61	-1.54	1.2×10^{-9}
3	D_3	515.19	38.38	212.29	5.09	66.07	4.05	1.09	-96.65	-1.35	1.2×10^{-9}
4	D_3	515.19	35.38	212.29	4.60	66.04	4.12	0.55	-96.70	-1.15	1.2×10^{-9}
5	D_3	515.19	32.38	212.29	4.11	66.00	4.18	0.01	-96.75	-0.96	1.3×10^{-9}
6	D_3	515.19	44.38	209.29	6.07	66.86	3.39	2.14	-96.96	-1.74	1.2×10^{-9}
7	D_3	515.19	41.38	209.29	5.58	66.82	3.46	1.61	-97.02	-1.54	1.2×10^{-9}
8	D_3	515.19	38.38	209.29	5.09	66.78	3.52	1.08	-97.06	-1.35	1.2×10^{-9}
9	D_3	515.19	35.38	209.29	4.60	66.75	3.59	0.54	-97.11	-1.16	1.2×10^{-9}
10	D_3	515.19	32.38	209.29	4.11	66.71	3.65	0.00	-97.16	-0.96	1.3×10^{-9}



(a) Joint angles calculated by MATLAB simulation
(b) Comparison with the teaching pendant display
(c) Physical insertion experiment

Figure 4.1: Import the simulation result from Matlab to real manipulator

4.3 Simulation Result In Reality

This following verifies whether the joint angles calculated through MATLAB inverse kinematics can be directly applied to the physical robotic arm to achieve the required wire-bending compensation. The purpose is to confirm that the pose compensation result obtained in the simulation is not only mathematically valid, but also executable in the real robotic system.

As shown in Figure 4.1(a), MATLAB calculates the six joint angles required for the compensated TCP pose. Figure 4.1(b) shows the physical robotic arm after applying the same joint angles obtained from the simulation. Under these identical angular conditions, Figure 4.1(c) demonstrates that the robotic arm can hold the bent wire and still align the wire tip with the socket. The wire can then be successfully inserted into the target socket, confirming that the MATLAB-based compensation model can be transferred to the physical experiment and satisfy the objective of wire angle correction.

4.4 Deviation Between Wire And Pin

In this wire insertion task, the theoretical maximum allowable error is significantly related to the wire and the socket. The tip diameter of the experimental wire is 0.7mm, denoted as R_{tip} , and the diameter of the insertion socket is 3mm, denoted as R_{socket} . For a single wire insertion to be successful, the outermost edge of the wire must not exceed the edge of the socket interface, which means:

$$D_{\text{max}} = \frac{R_{\text{socket}} - R_{\text{tip}}}{2} \quad (4.1)$$

This defines the absolute baseline for a successful wire insertion; D_{max} is 1.15mm, indicating that if the cumulative error causes the wire tip to deviate by more than 1.15mm from the center of the socket interface, even the tip will collide with the exterior of the socket surface, resulting in the absolute failure of that insertion attempt. This D_{max} is also referred to as the tolerance. Each wire and socket possesses its own manufacturing errors generated during production. However, these errors are negligibly small compared to the allowable tolerance; therefore, this study omits production errors from the analysis.

4.5 Unit Phase Alignment In Math

In Chapter 3, this study utilizes GeoGebra to calculate the rotation of two vectors in space, allowing the vector $\overrightarrow{B''C''}$ —formed by connecting the endpoints of these two rotated vectors—to align with the normal vector of the circular surface (which corresponds to the x -axis), as shown in the

figure. To mathematically prove that $\overrightarrow{B''C''}$ and the x -axis are aligned, the unit vector representation can be employed. Since a unit vector is defined as a vector with a length of 1 and is used exclusively to denote direction, proving that the unit vector of $\overrightarrow{B''C''}$ (expressed mathematically as $\hat{u}_{B''C''}$) is identical to the unit vector of the x -axis validates the alignment. The unit vector is obtained by dividing the vector by its own length, where the formula for the length of $\overrightarrow{B''C''}$ is:

$$\|\overrightarrow{B''C''}\| = \sqrt{\overrightarrow{B''C''}_x^2 + \overrightarrow{B''C''}_y^2 + \overrightarrow{B''C''}_z^2} \quad (4.2)$$

Applying $\|\overrightarrow{B''C''}\|$ to $\hat{u}_{B''C''}$, since $\hat{u}_{B''C''}$ is a spatial vector, it is expressed in terms of components across three directions:

$$\hat{u}_{B''C''} = \frac{\overrightarrow{B''C''}}{\|\overrightarrow{B''C''}\|} = \left(\frac{\overrightarrow{B''C''}_x}{\|\overrightarrow{B''C''}\|}, \frac{\overrightarrow{B''C''}_y}{\|\overrightarrow{B''C''}\|}, \frac{\overrightarrow{B''C''}_z}{\|\overrightarrow{B''C''}\|} \right) = (1, 0, 0) \quad (4.3)$$

When $\hat{u}_{B''C''}$ is identical to $(1, 0, 0)$, it indicates that $\overrightarrow{B''C''}$ shares the same direction as the x -axis, which is defined as being aligned.

4.6 Extreme Case In Simulation

After computing $\hat{u}_{B''C''}$, it is observed that due to the nature of spatial rotations, while this algorithm successfully counteracts the effects of large-angle deflections, a minor angular deviation persists after the rotation. Consequently, it cannot be stated that the vector is perfectly collinear and codirectional with the x -axis.

This subsection takes the most extreme angle from GeoGebra as an example. As shown in the figure 4.2, when the wire is bent at nearly 90° , the

angular deviation reaches an extreme condition. While the x -axis represents the forward direction toward the socket, $\overrightarrow{B''C''}$ possesses a non-zero value along the y -axis, indicating a slight offset. To investigate whether this offset exerts a significant impact by the time the wire moves to the front of the insertion socket, the slope can be used for the evaluation. The formula for calculating the slope is:

$$m_{xy} = \frac{\Delta y}{\Delta x} = \tan \theta \quad (4.4)$$

In other words, to calculate the angle θ :

$$\theta = \tan^{-1} m_{xy} \quad (4.5)$$

Substituting the values into the formula yields:

$$m_{xy} = \frac{\Delta y}{\Delta x} = \tan \theta = \frac{0.193}{15.318} = 0.0126 \quad (4.6)$$

$$\theta = \tan^{-1} m_{xy} = 0.72186^\circ \quad (4.7)$$

The physical significance of this slope indicates that for every 1mm advanced along the x -axis, a deflection of 0.0126mm occurs along the y -axis. Consequently, the total cumulative deflection upon reaching the front of the insertion socket is $0.0126 \times 15 = 0.189\text{mm}$. Alternatively, it can be stated that for every 1mm advanced, the y -axis introduces an additional angular deviation of 0.72186° . Since the allowable tolerance between the wire and the socket in the experimental environment is 1.15mm—which represents a 6.1-fold safety margin relative to the offset—this deviation still does not affect the experimental outcome. Furthermore, the slopes in the xz and yz planes are either 0 or infinite, implying that no distance-dependent angular drift occurs in those directions.

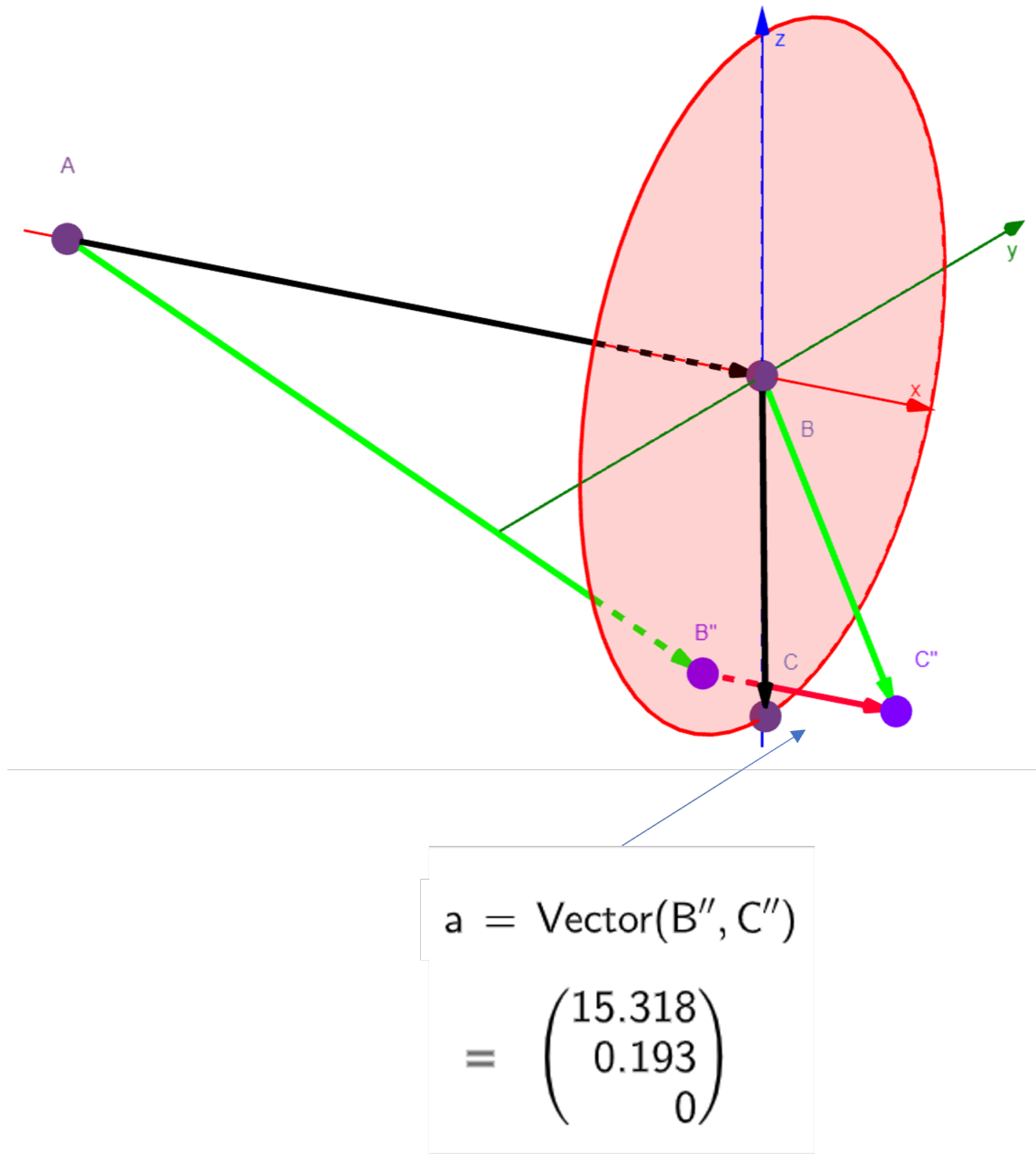


Figure 4.2: Extreme case show in Geogebra

Thus, the following conclusions can be drawn: Even under extreme conditions, θ never exceeds 1° . Given that the actual travel distance is relatively short, the wire tip is guaranteed not to deviate outside the boundaries of the insertion socket.

4.7 Transformation Matrix Integration

In Chapter 3, this study discusses the calculation methods and applications of the homogeneous transformation matrix. Through homogeneous transformation matrices, any position on the manipulator can be represented relative to the base coordinate system. This is not limited merely to the first six axes; after a gripper is integrated onto the arm, if the specifications of the gripper and its spatial distance from the sixth axis are known, then the center point of the gripper, and even the tip of the wire held by the gripper, can be incorporated into the equations as a seventh and eighth axis. This expands the homogeneous transformation matrix from the first axis through to the eighth axis.

To specifically describe the homogeneous transformation equation integrating both the seventh and eighth axes based on the vectors in GeoGebra prior to rotation, it can be written as:

$${}^6_8T = \begin{bmatrix} 1 & 0 & 0 & \overrightarrow{\mathbf{P}_{AB,x}} \\ 0 & 1 & 0 & \overrightarrow{\mathbf{P}_{AB,y}} \\ 0 & 0 & 1 & \overrightarrow{\mathbf{P}_{AB,z}} \\ 0 & 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} 1 & 0 & 0 & \overrightarrow{\mathbf{P}_{BC,x}} \\ 0 & 1 & 0 & \overrightarrow{\mathbf{P}_{BC,y}} \\ 0 & 0 & 1 & \overrightarrow{\mathbf{P}_{BC,z}} \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (4.8)$$

And after incorporating the rotation matrices, cross-referencing the equations from Chapter 3, it can be written as:

$${}^6_8T = \underbrace{\begin{bmatrix} R_y(-r) & * & R_z(s) & \overrightarrow{AB} \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{\text{The translation and rotation from point A to point B}} * \underbrace{\begin{bmatrix} R_y(r) & * & R_z(-s) & \overrightarrow{BC} \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{\text{The translation and rotation from point B to point C}} \quad (4.9)$$

By multiplying this with 1_6T ,

$${}^1_8T = {}^1_2T {}^2_3T \dots {}^6_7T {}^7_8T \quad (4.10)$$

The position of the eighth axis of the robotic arm—which is the wire tip—can be calculated relative to the physical manipulator base coordinate system.

Chapter 5 Conclusions and Future Works

This chapter consolidates the arguments and analyses presented in the preceding sections and provides a comprehensive summary of the findings.

5.1 Conclusions

This study successfully developed a model-oriented pose compensation framework that can potentially be adapted to different six-axis robotic manipulators and insertion products. By using the MDH convention to model the physical manipulator and combining the resulting kinematic model with the proposed TCP compensation algorithm, both the simulation and physical experiments demonstrated that the wire-tip orientation could be corrected through adjustments of the six joint angles without redefining the calibrated TCP stored in the manipulator controller. These results confirm the feasibility of using an external mathematical model to compensate for wire-orientation deviations in precision insertion tasks.

In the physical experiments, the wire was manually placed in the fixture before being grasped by the gripper. Therefore, the experimental outcome of each trial was classified solely according to whether the wire tip was successfully inserted into the corresponding socket. Under this criterion, the measured insertion success rate was approximately 53%, corresponding to a failure rate of approximately 47%. This result should not be interpreted as

reflecting only the accuracy of the proposed compensation model, because several uncontrolled factors also affected the physical insertion process. Only the upper-left socket was directly defined as the reference target in the model, while the coordinates of the remaining sockets were generated using fixed spatial offsets, as reflected in Table 4.3. Consequently, slight inclination of the socket platform could cause the actual socket locations to deviate from their calculated positions. Additional sources of failure included minor deformation of the wire tip caused by manual placement and grasping, residual TCP calibration errors, and the positioning accuracy of the physical manipulator. Accordingly, the experimental results demonstrate the feasibility of the proposed compensation framework while also indicating that improved fixture calibration, automated wire-pose measurement, and more accurate socket localization are required to achieve a higher and more stable insertion success rate.

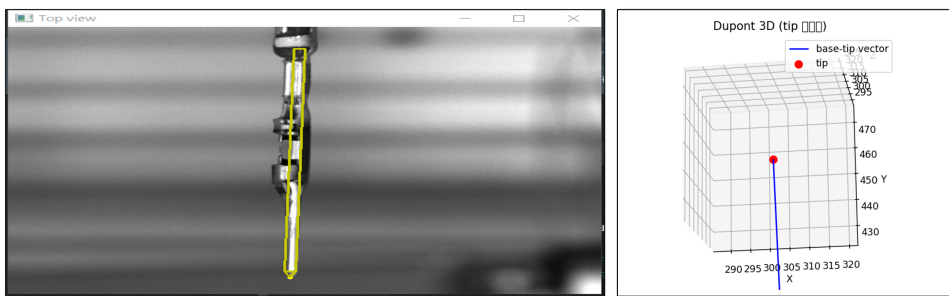
5.2 Future Work

The future prospects of this study are illustrated in Fig. 5.1. The current study assumes that the wire segment extending forward from the gripper fingers remains approximately rigid and does not undergo significant deformation. Therefore, the proposed compensation method mainly considers the angular deviation introduced by the fixed inclination of the gripper fingers, while deformation of the terminal wire segment itself is not explicitly modeled.

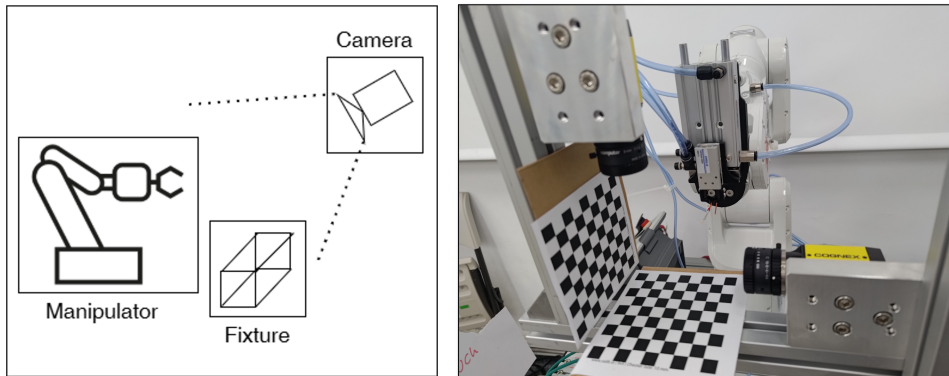
In future work, a vision system could be integrated to identify both the position and orientation of the wire tip and to estimate its actual bending

angle, thereby replacing the fixed-angle assumption adopted in this study. By combining visual measurements with the existing wire fixture and the MATLAB-based compensation model, the system could calculate an updated correction for each grasping or insertion cycle. This extension would allow the proposed method to handle cases in which the terminal wire segment is also flexible and the entire wire forms a curved or arc-shaped configuration. Even under such deformation, the local orientation of the wire-tip segment could be estimated and aligned with the normal direction of the target socket. In addition, robot-specific calibration and accuracy compensation data could be incorporated into the proposed kinematic model. Since individual commercial manipulators may exhibit slight differences in their geometric parameters and positioning characteristics even after calibration, introducing these robot-specific correction parameters into the model could reduce the discrepancy between the simulated and physical manipulator and further improve the accuracy of the calculated compensation results.

Furthermore, the proposed framework could be extended to support the simultaneous handling and insertion of multiple wires. By combining visual estimation of wire deformation, robot-specific calibration data, and the pose compensation algorithm developed in this study, a more comprehensive compensation framework could be established to account for errors originating from both the manipulated wire and the physical manipulator. Such an expansion would improve the adaptability of the system to different wire shapes, deformation conditions, manipulator characteristics, and assembly configurations, thereby enhancing its applicability to high-precision automated wire-insertion tasks in industrial environments.



(a) Camera identification of the wire tip pose (b) Import wire tip position to Matlab



(c) Import the model used in this research (d) Future application add in camera

Figure 5.1: Future work for this research

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